Do the gains from digital technology waves differ by income level? A systematic review, a pilot meta-regression and a macro-panel analysis

Quang-Vinh Dang*

Abstract

Background. Mobile telephony, the internet and, more recently, artificial intelligence (AI) are often expected to widen or narrow income gaps between countries, but the empirical cross-country evidence has not been synthesised across outcomes. **Methods.** We ran a systematic review following PRISMA 2020, searching OpenAlex, Crossref and NBER for 1995 to 2026 and, as a supplement, the first page of four Google Scholar queries, and assessed 132 full texts. We extracted effects on six outcome families, rated risk of bias with an adapted ROBINS-I tool, synthesised results narratively, and ran a pilot meta-regression of partial correlations for ICT-type adoption and output or productivity. For questions the studies cannot answer, we analysed World Bank data for 1995 to 2019 with fixed-effects panels, a staggered event study with not-yet-treated controls and local projections. **Results.** Eighty-eight studies met the criteria (71 on ICT, mobile, internet and broadband; 15 on AI; 2 on both). None was rated at low risk of bias (moderate 10, serious 67, critical 11). Most studies report a positive association between ICT-type adoption and output, but the pooled partial correlation across 22 studies is 0.16 (95% CI 0.09 to 0.23) with I^2 of 93%, and falls to between 0.04 and 0.09 after correcting for small-study bias. The moderator results are not stable: GMM estimates were 0.16 lower than others in a first run, and 0.07 higher (not significant) once studies found later were added, and restricting to studies with an identification strategy raises the pooled value instead of lowering it. Thirty-eight studies compare income groups or regions: nine find larger effects in richer economies, eleven in poorer ones, and four find no difference, mostly without a formal test. Returns are conditional on human capital, financial depth, governance and infrastructure. Evidence on AI comes from 17 studies, mostly short and using proxies. In the World Bank panel, mobile adoption was associated with faster growth in every income group under a same-period specification, most strongly in lower-middle income economies, but no income group showed a positive association under all four designs (same-period and predetermined panels, a staggered event study and local projections). **Conclusions.** The evidence does not establish that rich or poor economies gain more, and it cannot support causal statements. Credible identification for low- and middle-income economies is the main gap.

Keywords: digital technology; artificial intelligence; income level; systematic review; meta-regression.
JEL codes: O33, O47, O11, L96.

*British University Vietnam, Hung Yen, Vietnam. Email: vinh.dq4@buv.edu.vn. ORCID: 0000-0002-3877-8024.

1 Introduction

Whether new technologies narrow or widen income gaps between countries is a long-standing question, and it has returned with artificial intelligence. Early empirical work found sizeable output effects of telecommunications infrastructure [1]. Later work on the long-run diffusion of technologies finds that technologies arrive at similar times in rich and poor countries, but that the intensity of their use diverges [2, 3]. If so, the gains from a new technology should depend on a country's conditions at the time of adoption, and not only on whether the technology has arrived.

Evidence on this is scattered. Studies use different technologies (mobile subscriptions, internet users, composite ICT indices, ICT capital, robots, AI readiness indices), different outcomes (growth, productivity, employment, inequality, human development, emissions), different samples and different estimators. Individual studies often report one average effect and one or two interaction terms. A reader who wants to know whether a low-income economy can expect the same payoff as a high-income one has no single place to look, and a policymaker who reads one study can reach the opposite conclusion from one who reads another.

We therefore ask the following main question: *in empirical cross-country studies of digital and general-purpose technology waves, do the effects on output, structural change, labour markets, distribution, living standards and resource use differ by income level and by country characteristics, how large are the differences, and how credible is the evidence?* We keep studies of historical waves (ICT, mobile, internet, broadband, crypto-assets) separate from studies of AI, because AI evidence is younger and thinner. We also ask a set of subsidiary questions that the data we collected allow, and four questions that the published studies cannot answer, which we address with World Bank data.

The paper has three parts. The first is a systematic review that follows PRISMA 2020, with a structured extraction of effects, a quality rating of each study and a narrative synthesis across six outcome families. The second is a pilot meta-regression of effect sizes for the best-covered question, the association between ICT-type adoption and output or productivity, together with nine exploratory questions about measurement, study design and the conditions under which returns differ. The third is an original analysis of World Bank panel data that gives a first, transparent answer to four questions about the shape of the effect by adoption level, effects by income group under one design, timing around technology arrival, and differences inside low-income economies.

The contribution is a map of the evidence with an explicit quality rating of each study, a transparent count of where income-group comparisons exist and where they do not, an estimate of how much of the apparent positive association survives bias corrections, and a panel analysis whose limits are stated at each step. We are explicit about what the data cannot show. Data and code availability is described in the Declarations.

The rest of the paper is organised as follows. Section 2 sets out the conceptual background and the hypotheses. Section 3 describes the methods. Section 4 reports the review and the meta-regression, and Section 5 reports the panel analysis. Section 6 discusses the findings, and Sections 7 and 8 give the limitations and conclusions.

2 Conceptual background and hypotheses

2.1 Why the same technology could pay differently

Four mechanisms recur in the literature on information and communication technology and on automation, and each implies a reason why the gain from a technology might differ by income level.

Infrastructure and complementary investment. Telecommunications infrastructure raises output, and the effect may depend on how much infrastructure already exists, because network benefits grow with the size of the network [1]. Where electricity, roads and financial services are scarce, the same handset or connection may yield less. This implies that the gain depends on a country's stock of complementary assets, which is usually lower in poorer economies.

Information and market access. Mobile phones can reduce price dispersion and waste where markets are thin. In a study of South Indian fisheries, the arrival of mobile phones reduced price dispersion and waste and raised welfare [4], and a review of evidence from Africa describes how mobile phones can improve information flows in agricultural and other markets [5]. These mechanisms are strongest where information was previously scarce, which is more common in lower-income settings, and they imply large early gains at low adoption levels.

Skills and tasks. Computers substitute for routine tasks and complement non-routine tasks [7], and automation both displaces and creates tasks [8]. In Africa, the arrival of fast internet through submarine cables was associated with higher employment, and the effect was larger for more skilled workers [6]. The gains therefore depend on the skills of the workforce, which differ by income level.

Absorptive capacity. The ability to use external knowledge depends on prior related knowledge [9]. For technology adoption this means that returns depend on human capital, management capacity and institutions, again variables that differ systematically by income.

Two forces therefore pull in different directions. Catch-up and early information gains suggest that low-adoption, lower-income economies could gain more per unit of adoption. Complementary assets, skills and absorptive capacity suggest that higher-income economies could gain more. A third consideration is time: the long-run diffusion evidence suggests that gains may build over years, and a short window may miss them [3]. A review of the empirical evidence on information technology and economic performance, published early in the period we study, describes the findings as mixed [10]; it is not an estimating study and is not among our included studies.

2.2 Hypotheses and questions

These mechanisms lead to the following hypotheses, which structure the review and the panel analysis. They are stated so that the data can contradict them.

H1 ICT-type adoption is positively associated with output and productivity, but the size of the association is modest and varies strongly across studies.

H2 The association is larger where complements are stronger: higher human capital, deeper financial markets, better governance and better infrastructure.

H3 The association differs by income level, but its sign is not predicted, because catch-up and complement effects have opposite implications.

H4 Estimates from designs that address reverse causation are smaller than those from designs that do not.

H5 Gains depend on timing, appearing with a lag, so that studies measuring only contemporaneous effects understate them.

H6 Evidence on AI follows the same pattern as evidence on earlier technologies but is thinner and of lower quality.

3 Methods

3.1 Protocol and registration

The review follows PRISMA 2020 [11] and the structure of PRISMA-P [12]. The protocol (version 1.0, 2 October 2026) is available through the data and code link in the Declarations. It was intended to be registered on the Open Science Framework before the search, but registration had not been completed when the search was run. This is a deviation from good practice, and we log the amendments made after the search in Appendix H.

3.2 Eligibility criteria

We included quantitative empirical studies with their own estimation (panel regression, instrumental variables, growth accounting, event studies, structural estimation) that relate adoption, diffusion, investment or capability of ICT in general, mobile telephony, internet or broadband, crypto-assets or AI to at least one outcome in six families: (1) output and productivity, (2) structural change, (3) labour market, (4) poverty and distribution, (5) living standards beyond income (consumption, health, schooling, financial inclusion) and (6) resource cost (electricity use, carbon intensity). Historical-wave studies needed at least 10 economies or an explicit income-group comparison; AI studies needed at least 3 economies or an income-level comparison. Studies from 1995 onward in English were eligible. Purely conceptual papers, reviews and studies whose only outcome was adoption itself were not synthesised. Full criteria and exclusion codes are in Appendix D.

3.3 Information sources and search

We searched OpenAlex (a Boolean query restricted to the economics field), Crossref (96 technology-by-context queries) and the NBER working-paper series, covering 1995 to the search date of 2 October 2026. The search blocks and the OpenAlex query are in Appendix C. arXiv was attempted but did not respond in usable time. Scopus, Web of Science and EconLit were not searched, because they need institutional access. After removing duplicates by DOI and by normalised title and year, 3,612 records remained.

A blind re-screening of the retrieved records suggested that the search had missed relevant studies (Section 4.5). A supplementary check in Google Scholar, where the author ran four of 15 planned queries (G1, G2, G4 and G7, listed in Appendix C) and passed on the first page

of 10 results for each, found that most results were not among the 3,612 records (9 of 10 for G1, 6 of 10 for G2, 7 of 10 for G4 and 7 of 10 for G7); of two further results that were among them, one (Yin and Choi 2023) had been removed by the automated stage-1 rule and the other (Gonzales 2023) had been judged includable but never read for lack of a full text. These records are handled outside the main flow (Table 1). Eleven planned queries have not been run, and the search as a whole should therefore be regarded as incomplete.

3.4 Screening and full-text access

Screening was done by one assisted reviewer. Automated rules removed records with no eligible technology term, no outcome term or no multi-economy cue (stage 1). Title and abstract screening then applied the eligibility table, with four holding categories for records that were not eligible but useful (context on diffusion, environmental-only outcomes, reviews for citation search and supplementary micro evidence). No human second screener was available. Instead, a separate model run, which saw only the criteria and each record's title and abstract, independently re-screened a random sample of 340 records (100 of those removed by the automated rules, 120 of those removed at title level and 120 of those assessed at abstract level; seed 20261004). Records the first screener excluded and the second would include were adjudicated by a third blind reading of the abstract. We report agreement and an estimate of missed records in the Results.

Full texts were obtained from open-access sources, repositories and the author. Records without a free full text, and those the author judged to come from unreliable venues after visiting the journal site, were not read and are not used as evidence. This creates a selection mechanism that favours open-access studies; Section 7 discusses it. Three studies were added late (71, 278 and 341) after the author supplied them; one further study (89) was supplied and excluded. After the blind re-screening, five more records it flagged were obtained and read: four are included (448, 755, 891 and 9001) and one (474) was excluded. The twenty studies from these Google Scholar pages that the author could download (9101 to 9106, 9201 to 9204, 9301 to 9305, 9401 to 9404 and 1580) were read, extracted and checked in the same way as the others; three of them (9202, five countries; 9301, a mathematical model without estimation; and 9403, estimated only on a simulated panel) were excluded, and one (9401) was kept on the strength of its generic ICT index alone, because the paper does not measure AI. Three further results of the G1 page (Chavula 2013, Torero et al. 2006 and a 2026 preprint) could not be obtained.

3.5 Data extraction

For each included study we extracted technology, exposure measure, sample, estimator, identification strategy, results by outcome family with printed estimates and uncertainty, heterogeneity findings, and group-specific results, using a fixed form (Appendix D). Extraction was done by a language-model agent working from converted full texts, with a locator and a short verbatim quote for every result. An automatic check matched 515 of 756 quotes verbatim across all 132 records read (the remainder are affected by table layout and paraphrase).

3.6 Second-pass check

A second, separate model run then re-read each of the 88 included records and each of 317 regression estimates against the text and classified every item as confirmed, corrected or unverifiable. This catches transcription and reading errors. It is not an independent check: the same family of model reads the same machine-converted text, and garbled tables stay garbled. Appendix F summarises the second pass by study. In addition, a separate model run independently re-extracted key fields (number of economies, years, estimator class, identification, outcome families, headline direction on output, income-group comparison, low-income coverage and overall risk of bias) for 16 randomly chosen included studies (drawn before the Google Scholar studies were added) without seeing the original extraction, and we compare the two. Finally, the author checked several randomly selected samples of the extracted data by hand against the source papers. The sizes of these samples and the corrections made were not logged, so this check is described here but not quantified.

3.7 Risk of bias

Risk of bias used a six-domain tool adapted from ROBINS-I [14]: confounding and reverse causation, exposure measurement, selection of countries and years, outcome measurement, specification and researcher degrees of freedom, and selective reporting. The overall rating was low, moderate, serious or critical, with the definitions in Appendix D. One rater made the first-pass ratings; the second pass offered an opinion on each, which we report without changing the ratings.

3.8 Narrative synthesis

We synthesised results narratively, following the SWiM reporting guideline [13], by outcome family and by technology stream. Studies were grouped by outcome family; within each family we tabulated the direction of the results, the estimator, the sample and the risk of bias (Appendix B). Counts of study directions are descriptive; they are not weighted by sample size or quality and are not effect sizes.

3.9 Pilot meta-regression

For ICT-type adoption and output or productivity we ran a pilot meta-regression in the tradition of Stanley and Doucouliagos [15]. Each estimate was converted to a partial correlation coefficient, $PCC = t/\sqrt{t^2 + df}$, with standard error $\sqrt{(1 - PCC^2)/df}$ and df equal to the number of observations minus 10 regressors (a sensitivity check uses 0, 5 and 20). The t-statistic came from the printed coefficient and standard error, or from a printed t-statistic. Estimates reported with p-values or intervals only were not converted. We excluded interaction terms, crypto and AI exposures, estimates without a usable number of observations, and five estimates judged not poolable on the second pass (a fintech-credit exposure, a growth outcome mixed with level outcomes, and a TFP index in levels). We pooled one headline estimate per paper by random effects [17], ran a weighted moderator regression with standard errors clustered by paper, and tested for publication selection with the funnel-asymmetry and precision-effect tests (FAT-PET) and

the precision-effect estimate with standard error (PEESE) [16]. With 24 papers, the clustered standard errors are optimistic and p-values are indicative.

3.10 Additional exploratory questions

Beyond the main question we specified nine further questions and their analyses before running them, all using the data already collected: whether the pooled association depends on the measurement of the technology (a composite index versus a single indicator), on the outcome form (growth versus level) and on the sample period (added one at a time to the meta-regression); whether weaker designs are more common where low-income economies are sampled; whether study quality goes with the reported direction; which enabling conditions authors test and in which direction (170 heterogeneity entries classified by one model reader, with a second reader on a random 20% sample); the thresholds stated; an evidence-gap map by outcome family and income group; and a comparison of the AI and historical-wave streams. These analyses are exploratory, are not corrected for multiple testing, and should be read as hypotheses.

3.11 Macro-panel analysis

Four questions (the shape of the effect by adoption level, effects by income group under one design, timing around technology arrival, and differences inside low-income economies) cannot be answered from the included studies. We therefore specified, before downloading the data, an analysis of public World Bank World Development Indicators for 1995 to 2019, and later amended the event-study design after the first design failed its own checks. The variables and their World Bank codes are in Appendix G.

Fixed-effects panels (A1 to A3, A6). In five-year periods (1995–2000, 2000–05, 2005–10, 2010–15 and 2015–19), annual growth of GDP per capita was regressed on the change in mobile subscriptions per 100 people over the same period (A1) or the previous period (A2), with initial log income, initial gross secondary enrolment, the change in trade openness and period effects, separately by current World Bank income group, with standard errors clustered by country. A3 splits the change by baseline adoption level, and A6 adds enabling conditions inside low and lower-middle income economies, one moderator at a time.

Event study (A4b). The takeoff year is the first year mobile subscriptions reach 10 per 100 people among economies below that level in 1995. For each cohort of economies with the same takeoff year and each horizon from five years before to ten years after, the effect compares the change in log GDP per capita from the year before takeoff with the change in economies not yet treated at that horizon, after adjusting for initial income and earlier growth by outcome regression, in the manner of Callaway and Sant'Anna [18]. Estimates are averaged across cohorts, weighted by the number of treated economies, and intervals come from a bootstrap over economies. Effects at horizons before takeoff are placebo tests, and a subgroup is interpreted only if the interval of their mean includes zero. We use not-yet-treated controls because standard two-way fixed-effects event studies can be biased when treatment is staggered and effects vary over time [19, 20].

Local projections (A7). For a design without a binary treatment, we estimated local projections [21] on the annual panel: cumulative growth of GDP per capita from year t to $t + h$ on

the change in mobile subscriptions over the previous three years, with country and year fixed effects, controls for earlier growth and initial income, and separate coefficients by income group.

All of these are associations. We have no instrument, so we do not claim a causal effect, and cost-effectiveness cannot be examined with these data.

4 Results of the review and the meta-regression

4.1 Study selection

Figure 1 and Table 1 give the flow. Of 247 records sent to full-text assessment, 106 full texts were obtained and read, together with one review, five records outside the 247 that the blind re-screening flagged (Section 4.5) and twenty records found through Google Scholar. After the second pass, 88 studies are included, 43 were excluded at full text (14 for too few economies, 5 for an ineligible exposure, 1 for an ineligible outcome, 23 for being conceptual or lacking their own estimation) and one was kept as context. Four studies (298, 474, 878 and 1495) were moved from include to exclude after the second pass on eligibility grounds; this should be confirmed by a human reviewer. About 140 of the 247 records have not been read because no free full text was available.

Table 1: Study selection (counts as recorded by the author).

Stage	n
Records identified (OpenAlex 1,840; Crossref 1,285; NBER 579; before cross-source deduplication 3,704)	3,704
Duplicates removed across sources	92
Records after deduplication	3,612
Google Scholar, queries G1, G2, G4 and G7, first page each (supplementary): results / not in the main search / obtained and read	40 / 29 / 20
Excluded by automated stage-1 rules	1,545
Read at title level / excluded at title level	2,067 / 1,611
Assessed at abstract level / excluded at abstract level	456 / 121
Held as duplicates, context, environmental-only, reviews, supplementary	88
Sent to full-text assessment	247
Full texts read (106 of the 247, one review, five records added after the re-screening and twenty from Google Scholar)	132
Excluded at full text	43
Kept as context only	1
Included	**88**

4.2 Characteristics of included studies

The 88 included studies were published between 2003 and 2026, and 70 of them from 2020 onward (Figure 2). Seventy-one study ICT, mobile, internet or broadband; 15 study AI; 2 cover both. Most use country-year panels (70), with a few at industry-country level. In a rough classification of the main estimator, 35 use system or difference GMM, 32 use fixed-effects, random-effects or pooled OLS, 13 use ARDL, cointegration or heterogeneous-panel estimators and 8 use growth accounting, data envelopment or other methods. Forty-two studies have no identification strategy beyond controls and fixed effects. Twenty-eight list at least one low-

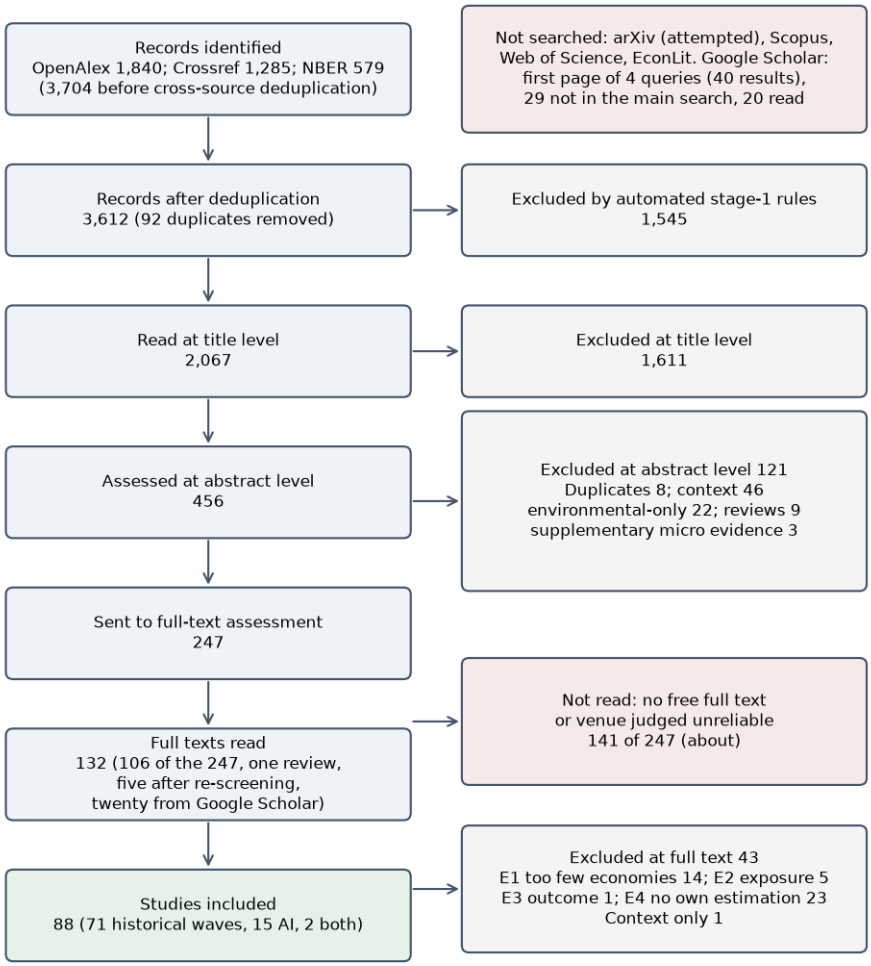


Figure 1: PRISMA 2020 flow diagram.

income economy; studies that do not state their income coverage are counted as not listing one. Outcome families, with a study able to contribute to several, are: output and productivity 53, labour market 15, living standards beyond income 16, poverty and distribution 13, structural change 6, resource cost 6. Appendix A lists all studies and Appendix B gives the results by family.

4.3 Risk of bias

No study was rated at low risk of bias. The first-pass ratings were moderate for 10 studies, serious for 67 and critical for 11 (Figure 3). In the second pass, the separate reader agreed with the overall rating for 77 of 88 studies, found seven too harsh (268, 302, 323, 408, 707, 1469 and 9401) and four too lenient (57, 957, 1031 and 9204). We did not change any rating, and these differences should be settled by a human reviewer. The commonest weaknesses are adoption treated as exogenous, indices whose scope does not match the technology, and many specifications with no stated preferred one.

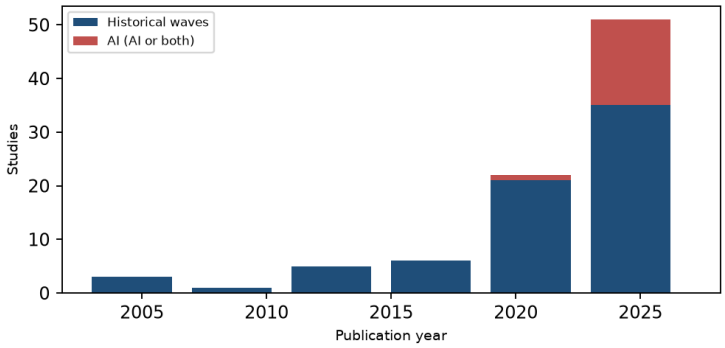


Figure 2: Included studies by publication year and stream.

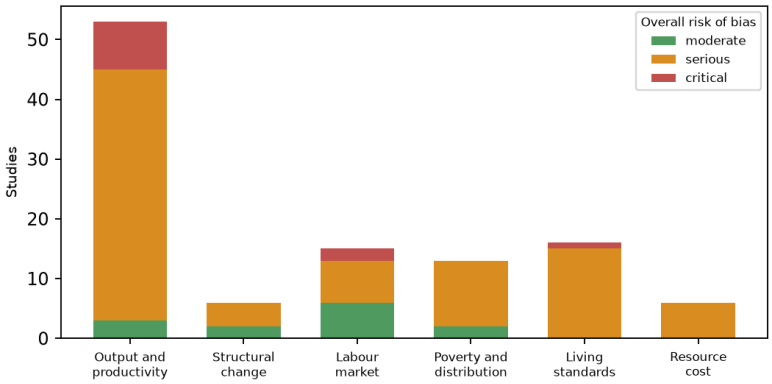


Figure 3: Overall risk of bias by outcome family (a study counts in each family it reports).

4.4 Accuracy of the extraction

Across 531 result entries in the 88 re-read records, 492 were confirmed, 34 corrected and 5 could not be verified; 18 of 170 heterogeneity entries were corrected. Of 317 regression estimates, 311 matched the printed tables and six needed correction: one income label and the five estimates of study 9103, which had been read from the wrong cell of a panel-VAR table (rows are equations and columns lagged regressors, as the paper's own Granger tests show, whereas its text reads the table the other way). In that study the extraction had followed the paper's text, so the second pass caught an error that originated in the source and that the first pass had copied. Corrections were mostly misread labels, counts, and claims that the source text does not make. They were applied or flagged in the data files. Because both passes used the same family of model reading the same machine-converted text, the author's hand check against the source papers is therefore the reference; as noted in Section 3, its sample sizes and corrections were not logged.

4.5 Independent cross-checks of screening and extraction

Screening. The blind second screener agreed with the first on whether to send a record to full text for 79% of the 120 records in the abstract-stage sample (Cohen's kappa 0.59 when its "unsure" counts as include and 0.54 when it does not). Its disagreements were mostly cases where it was more inclusive: of the 72 records the first screener did not send forward, it would have included

or flagged 20. Of the 8 sampled records the first screener set aside as environmental-outcome-only (protocol amendment 2), the second screener would have sent 7 to full text. Carbon and electricity outcomes are in outcome family 6, so such records meet the eligibility criteria; setting them aside was a prioritisation choice, and it explains in part why only four included studies report resource-cost outcomes.

Recall. Among the records excluded earlier, the second screener would have included or flagged 7 of 100 (automated rules), 15 of 120 (title level) and 8 of 120 (abstract level, coded exclusions). After a third blind reading of the abstracts, 2 of 100, 10 of 120 and 4 of 120 met or probably met the criteria. Extrapolated, this implies roughly 31 (range 9 to 108), 134 (74 to 236) and 15 (6 to 38) missed records at the three stages, on the order of 180 in total (range about 90 to 380). These are small samples read by model readers, and 9 of the 30 adjudicated records had no abstract, so the figure is a warning about recall and not a count. Later first-page checks of four Google Scholar queries found 29 of 40 results missing from the 3,612 records (and two more removed by the stage-1 rule or never read), and seventeen of the twenty studies read from them were included, so the main search had poor recall for this question and the estimate above is a lower bound. The missed records are typically AI or digital-economy panels with carbon or energy outcomes, small AI panels, and financial-inclusion studies in Africa and the Arab world. Adding two such studies reversed the sign of some moderator estimates in the meta-regression (Section 4), so conclusions that rest on moderators are fragile, and the evidence base is smaller than the literature.

Extraction. A blind re-extraction of key fields for 16 random included studies agreed with the original on the outcome families (16 of 16), start and end year (15 of 16 each), the presence of an identification strategy (14 of 16), the direction of the headline output result (14 of 16), the number of economies within 5% (13 of 16), whether the estimator is GMM (13 of 16), the four-way estimator class (12 of 16), whether income groups are compared (12 of 16), and the overall risk-of-bias rating (11 of 16). Low-income coverage agreed in all 5 studies where the blind reader could judge it. Disagreements arise from papers with several estimators, differing sample counts in the text and the tables, and the subjective risk-of-bias scale: where the two differ on risk of bias, the blind reader is more lenient in four of five cases, in line with the second-pass opinions. We therefore treat the estimator classes and risk-of-bias ratings as approximate, and the count of studies that list a low-income economy (28) as uncertain by a few studies.

4.6 Output and productivity

Fifty-three studies cover output or productivity (7 on AI). Most report a positive association, but the sign is not uniform: current and lagged mobile effects have opposite signs in one 41-economy GMM study (study 10); ten high-income countries show no significant effect of any ICT indicator (study 1448); and in 27 EU states the digitalisation coefficient loses significance once economic freedom is added (study 988). In 11 MENA economies, CS-ARDL estimates of an ICT index on GDP per capita are 0.05 (long run) and 0.06 (short run), both significant only at 10%, while the dynamic common correlated effects estimate is 0.30; the paper's own causality tests show feedback in both directions (study 71). A digital economy index lowers green total factor productivity across 40 Belt and Road economies (study 245), although that outcome sits closer to resource cost.

The studies found later through Google Scholar are no more uniform. Internet use is negatively associated with growth of GDP per capita across 78 economies, significantly so only in the high-income group (-0.021, against -0.012 pooled and $+0.006$, not significant, in lower-middle income economies; study 9102). An instrumental-variable study of 123 economies finds positive elasticities of GDP per capita with respect to an ICT index in all three income groups, largest in middle-income economies (0.42), then low-income (0.16) and high-income (0.09) (study 9105). A study of the ten ASEAN economies, none of them low-income, finds fixed broadband positively and mobile broadband negatively associated with GDP per capita (study 9101). A study of 85 low- and high-income economies finds small positive associations of internet users and fixed telephones with GDP per capita in both groups and a negative association of secure internet servers in low-income economies (study 9203). In 182 regions of 27 EU countries, internet use and e-government are mostly positively associated with total factor productivity, with some negative interactions (study 9302). A study of 45 developed and developing countries finds that the output elasticity of IT is similar across the two groups, with a larger complementarity with tertiary education in developing countries (study 9305). A study of 42 sub-Saharan economies reports that an ICT index, a technology-readiness index and the share of technology imports are positively associated with growth of GDP per capita, although the paper never measures AI, its stated subject, and is rated critical (study 9401); a composite digital index is positively associated with output per worker and with GNI per capita in 25 economies (study 9402). Studies of sub-Saharan Africa (study 9304) and of transition economies (study 9303) find thresholds, an inverted U above about 100 mobile subscriptions per 100 people in the first and a level-dependent interaction between IT investment and staff in the second, the latter rated critical. A subnational study of about 34,000 districts in 120 countries finds that expansion of mobile coverage raises nightlight growth, a proxy for local activity, which the author converts to an implied GDP elasticity of 0.018 to 0.023 using an external nightlights-GDP elasticity; the 2G effect is concentrated in developing countries and effects are larger where fixed-line telephone penetration was low (study 9106). Two panel-VAR studies (9103 and 9104) report only p-values, so they could not enter the meta-regression, and in both the text disagrees with the table. In study 9104 the abstract reports a positive effect of ICT on growth, but the printed table shows negative coefficients of lagged ICT on real GDP in the full sample, sub-Saharan Africa and Latin America and a positive one only in the Middle East and North Africa; read with the layout the paper's own Granger tests imply, study 9103 shows positive effects of lagged ICT on real GDP in all four income groups, although the outcome is a level in billions of dollars and the coefficients are not comparable across groups.

Interaction terms are the most repeated finding: ICT pays more where human capital is higher in two Asian GMM studies (302 and 323, which share authors, panel and index), a developing-country productivity study, and in Africa, where the effect of the digital economy is negative below a human-capital index of about 2.35 and the average country is below it. Financial development, trade openness, governance, electricity and the diffusion level of ICT also appear as moderators. These are conditional associations from single-equation panels and do not show that raising a moderator would change the return. A non-parametric study of 24 OECD economies finds that ICT capital change has no contemporaneous effect on the production frontier and a positive effect five years later, which supports the view that timing matters (H5). In the five AI studies, developed economies show about twice the productivity effect of emerging economies (0.24 against 0.11) in a 10-economy comparison rated at critical risk of bias; in 63

low and lower-middle income economies, AI-related imports are associated with higher labour productivity, more so where connectivity, electricity and financial inclusion are better; and two further studies (a sub-Saharan firm-level study and a three-country generative-AI survey) are rated critical and cannot be relied on. A study of nine economies (the G7, China and Korea) finds mostly null or sign-changing associations between an AI proxy and GDP per capita, and the proxy in some models is an ICT export share (study 891).

4.7 Structural change

Six studies cover structural change. Internet penetration is associated with faster structural change in a 51-economy panel, more so in low and lower-middle income and agrarian economies than in upper-middle and high-income ones, and more so where forward value-chain linkages are higher; manufacturing employment falls in South Asia and is unchanged in sub-Saharan Africa. A 3G coverage study in 14 developing countries finds jobs created in services and non-farm self-employment but no shift of labour away from agriculture. ICT is positive for industry and services value added but not for agriculture in a 26-economy PMG study, although columns 2 to 4 of its Tables 10 and 11 are identical across industry and services, so that paper needs a human check. A study of about 100 developing economies finds that ICT lowers export concentration, with a larger effect in the low and lower-middle income group (-0.155 on the concentration index) than in the high and upper-middle group (-0.020), but no formal test of the difference. A panel VAR of 171 economies (study 9103) finds, on a reading consistent with its table layout, a weakly positive association of lagged ICT with the manufacturing share in every income group, whereas its authors read the table as a large negative effect of ICT on industrialisation. The evidence base here is still small.

4.8 Labour market

Fourteen studies cover labour markets, five of them on AI. In advanced economies ICT is linked to routine-task decline and polarisation, and capital-skill complementarity raises skill premia. In 27 European countries ICT capital creates more jobs where the share of high-skilled jobs is lower. No included study measures routinisation in low-income economies. Third-generation mobile coverage raises women's labour-force participation but not men's in 14 low- and middle-income countries, in one of the few studies that uses an instrument for developing-country labour outcomes. AI exposure is positively associated with employment growth, concentrated in high-education occupations and younger workers, in European samples; in Southeast Asia the pattern differs by country, though the country-level estimates could not be checked in the source figure. Across 29 countries, the AI index effect on employment declines with labour productivity level. A 78-economy readiness-index study is rated critical. The pattern across studies is that AI complements skills and displaces the rest, and the evidence on low-income economies is nil.

4.9 Poverty and distribution

Twelve studies cover poverty and distribution, two on AI. Results split. In 19 G20 economies, internet use is associated with a lower Gini coefficient in the middle-income group (-0.020) and a higher one in the high-income group ($+0.047$), with split samples and no formal test of the

difference (study 9201). A threshold model for 45 developing countries finds that ICT raises the Gini coefficient below a digital-maturity score of 0.48 and lowers it above, an inverted U. In 48 sub-Saharan African economies ICT is positively associated with a composite inclusive-growth index (0.03 to 0.15 across specifications), with trade openness negative on its own and offset by ICT in the interaction, though the paper is inconsistent about its estimator. A study of poverty and inequality finds that the effect of inequality on poverty reverses above a technology threshold, mobile banking is mostly null or weakly negative on average inclusive growth with effects at the tails of the distribution, and robots raise carbon inequality between countries, less so in open and well-connected economies. The consistent message is that effects depend on how far a country has gone: below some level digitalisation widens the gap, above it narrows it. The thresholds are estimated in single studies with large uncertainty and have not been replicated.

4.10 Living standards beyond income and resource cost

Sixteen studies cover living standards. Mobile subscriptions are positively associated with the Human Development Index in high-income and lower-middle-income groups and not in low-income countries in a large panel. ICT slopes on the index are positive in South Asia, sub-Saharan Africa and Latin America. A panel VAR of 73 economies finds lagged ICT positively associated with the index in the full sample, sub-Saharan Africa and Latin America and negatively in the Middle East and North Africa (study 9104). In a system-GMM study by income group, the ICT coefficient on the index is larger in low-income economies (0.054) than in lower-middle income ones (0.006) and not significant in the two richer groups, but the low-income and upper-middle estimates are degenerate and the sample counts in the tables do not reconcile (study 9204). In low and lower-middle income countries ICT narrows gender education gaps up to about 100 mobile subscriptions per 100 people, after which the gap widens again. Internet raises financial inclusion more where governance is better in sub-Saharan Africa, and mobile helps only where internet access is adequate. An AI development index raises human-capital-based educational development in developed economies and not in developing ones, with a sharp pandemic-period jump. Mobile phone subscriptions in 88 countries are associated with higher brain cancer mortality at long lags (15 to 20 years) in a difference-in-differences style panel, with similar marginal effects in OECD and non-OECD countries; the authors disclaim causality and the main result disappears without country-specific trends (study 9001). In 45 Sub-Saharan African economies, mobile phone use is associated with higher banking inclusion, whereas the text's claim of an internet effect conflicts with a non-significant coefficient in the main table, and the paper is inconsistent about its estimator and sample (study 448).

Six studies cover resource cost, four on AI. Digitisation is associated with higher carbon productivity, and the effect falls with income (0.085 for high-income, 0.058 for middle-income, 0.041 for low-income countries), while a digital-economy index lowers green total factor productivity in Belt and Road economies. Two AI studies of small samples, ASEAN-5 and a group of nine advanced economies, find that AI investment is associated with lower emissions only in a high-FDI regime that rests on one country (study 755) and with energy use in ways that change sign across specifications (study 891). Six studies are too few to generalise.

4.11 Do gains differ by income level?

Thirty-eight studies compare income groups or regions, formally or descriptively (Table 2). Nine find larger effects in richer or OECD economies, eleven in poorer or less productive ones, and four find no clear difference; one finds a negative effect in all groups. Several of the first group have no test of the difference, and the studies finding larger effects in poorer economies include one single cross-section rated critical and one (9102) in which the coefficient is negative everywhere and merely less negative in poorer groups. The remaining thirteen studies compare groups without a directional finding (mixed by technology, coefficients not comparable across groups, or regional splits only), and study 268 appears in both of the first two groups because its direct and moderating effects point in different directions. We therefore treat the evidence as inconclusive. Studies of richer economies mostly use OLS or fixed effects, test the difference informally, and rarely include low-income economies, so the lean toward larger measured gains in richer samples may reflect study design.

Table 2: Studies that compare income groups or regions (study IDs in Appendix A).

Finding	Studies	Notes
Larger in richer or OECD	88, 138, 162, 268 (direct effect), 276, 535, 750, 957, 1580	Several lack a test of the difference; 535 rated critical and its figures look illustrative; 957 compares OECD with MENA
Larger in poorer or less productive	268 (informality moderation), 341, 407, 707, 952, 1192, 9102, 9105, 9106, 9201, 9204	407 compares Central and Eastern with Western Europe; 1192 is a single cross-section rated critical; 952 flips sign between pooled and subgroup models; 9102 is negative everywhere and less negative in lower-middle income economies; 9105 is largest for middle income on an ICT index but largest for high income on internet and fixed broadband; 9106 compares developing with developed countries (nightlights); 9201 finds that digitalisation raises the Gini coefficient in high-income and lowers it in middle-income G20 economies; 9204 has degenerate estimates in two columns
No clear difference	29, 320, 879, 9001	879 compares low with lower-middle income only; 9001 reports similar marginal effects for OECD and non-OECD countries
Negative in all groups	245	Green total factor productivity outcome
No directional finding	9101, 9103, 9104, 9203, 9305 (and others)	Mixed by technology (9101); levels not comparable across groups (9103); regions, not income groups (9104)

4.12 Evidence on AI

The 17 studies on AI (15 AI only, 2 both streams) are mostly short panels of 3 to 122 economies, typically starting in 2012 or later, with exposure measured by occupational exposure indices, national readiness indices, patent-based measures or industrial robot density. None measures realised use of generative AI at country level. Where AI studies show an effect, it tends to

be larger in richer economies and in high-skill occupations. The evidence from low-income economies is one productivity study of 63 low and lower-middle income economies and two studies rated critical. Two further studies add little to that picture: a five-economy panel of BRICS countries finds that governance moderates the association of AI capital investment with growth and poverty, with some interactions positive and some negative (study 9404), and a 122-economy panel finds that AI patents raise growth in advanced economies and not in less advanced ones (study 1580). AI cannot be compared with earlier waves on equal terms.

4.13 Pilot meta-regression: ICT-type adoption and output

Of 318 estimates from 38 papers, 148 from 24 papers were usable (71 reported p-values only, among them six studies found through Google Scholar: 9101, 9103, 9104, 9303, 9304 and 9402). For the 22 papers with a convertible headline estimate (Figure 4), the random-effects pooled partial correlation is 0.161 (95% CI 0.093 to 0.228), with $\tau^2 = 0.023$, $I^2 = 93\%$ and a 95% prediction interval from -0.144 to 0.465. A partial correlation of 0.16 corresponds to about 2.6% of residual variance. Of the headline estimates, 86% are positive and 68% are positive and statistically significant. A first run of the pilot, before studies found through Google Scholar were added, gave 0.189 (95% CI 0.104 to 0.274) from 16 papers, so the pooled value is stable; the moderator results, below, are not.

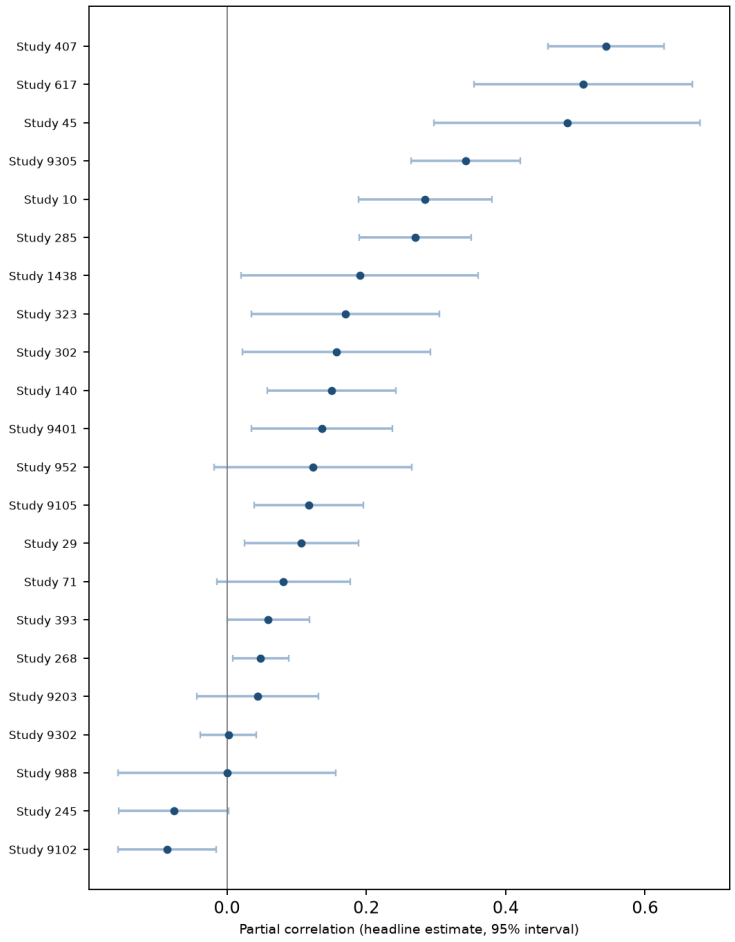


Figure 4: Headline partial correlations by study (95% intervals).

Table 3: Moderator meta-regression of partial correlations (148 estimates, 24 papers, standard errors clustered by paper).

Moderator	Coefficient	Clustered SE	p
Constant	0.075	0.048	0.12
Low- or middle-income sample	0.079	0.073	0.28
System or difference GMM	0.069	0.077	0.37
Mobile (relative to general ICT)	0.047	0.043	0.27
Internet or broadband (relative to general ICT)	−0.082	0.056	0.15

Table 3 shows no moderator that is distinguishable from zero. In the first run (103 estimates, 18 papers), GMM estimates were 0.16 lower in partial correlation than other methods ($p < 0.01$) and internet estimates 0.14 higher ($p = 0.09$). After adding the 45 usable estimates of six studies found through Google Scholar (9102, 9105, 9203, 9302, 9305 and 9401), the GMM coefficient is +0.07 ($p = 0.37$) and the internet coefficient −0.08 ($p = 0.15$). A moderator whose sign changes when a few studies are added is not a finding, and the earlier reading that GMM estimates are lower, which looked like support for reverse causation inflating simpler estimates, cannot be maintained; H4 therefore receives no support from the meta-regression. The low- or middle-income indicator is not significant either; 10 papers (33 estimates) come from such samples, against 21 papers and 115 estimates for high-income or mixed samples (mean partial correlation 0.146 against 0.114, unweighted; in the first run the gap was the other way, 0.031 against 0.123). The FAT-PET slope on the standard error is now 1.60 ($p = 0.11$) with a bias-corrected intercept of 0.036 ($p = 0.47$), and the PEESE intercept is 0.086 ($p = 0.03$); in the first run the slope was 1.91 ($p = 0.06$) and the corrected values 0.005 and 0.064. The evidence of selective reporting is therefore weak, but the corrected effects (0.04 to 0.09) remain well below the uncorrected 0.16 (Figure 5). The pooled estimate is stable to the degrees-of-freedom assumption (0.158 to 0.163 for 0, 5 and 20 regressors).

Sensitivity analyses. Dropping each study in turn moves the pooled partial correlation between 0.138 and 0.173, so no single study drives the result. Excluding the studies rated critical gives 0.146 (20 papers, 95% CI 0.077 to 0.215), and excluding one of the two studies that share authors, panel and index (323) gives 0.160. Only two papers with a usable headline estimate were rated at moderate risk of bias, so a restriction to moderate-risk studies is not possible. Restricting to the 13 studies whose design includes an identification strategy beyond controls (internal or external instruments, lagged regressors or a timing design) gives a *higher* pooled value of 0.184 (95% CI 0.107 to 0.261, $I^2 = 89\%$), not a lower one. Neither the estimate-level GMM indicator nor the study-level restriction therefore shows that better identification lowers the association. Units differ across studies, and GMM estimates in these studies are often short-run effects of levels on growth rates, which may account for part of the differences.

4.14 Exploratory analyses of additional questions

Measurement, outcome form and period. Adding single variables to the meta-regression, composite-index exposures show a partial correlation 0.04 higher than single indicators (SE 0.08, $p = 0.62$) and growth-rate outcomes 0.04 lower than level outcomes (SE 0.09, $p = 0.62$); neither is distinguishable from zero, and both rest on between-paper contrasts because few papers contribute both kinds. Later samples show lower partial correlations (−0.015 per year of sample midyear,

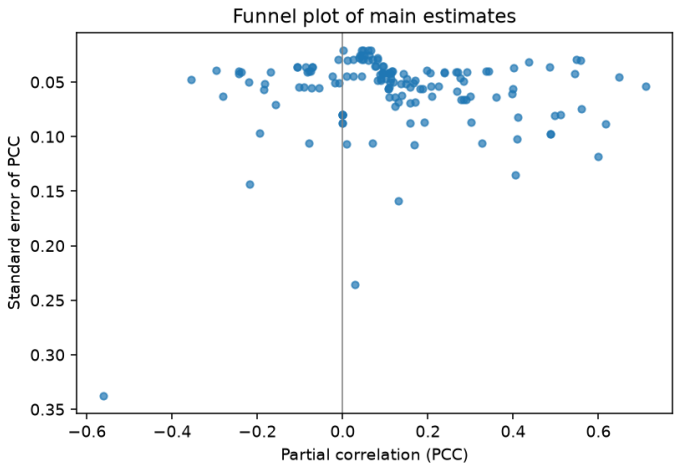


Figure 5: Funnel plot of main estimates (partial correlation against its standard error).

SE 0.007, $p = 0.03$), which would be about -0.15 per decade. This is the only one of the exploratory tests below 0.05, it moved from $p = 0.13$ to 0.03 as studies were added, it is partly confounded with the shift to GMM in recent studies, and with eight exploratory tests and no correction for multiplicity it is a hypothesis about diminishing returns, not a finding.

Design and income coverage. Studies that list a low-income economy were not more likely to lack an identification strategy (10 of 28, 36%, against 32 of 60, 53%; Fisher $p = 0.17$) or to be rated at higher risk of bias (moderate risk: 3 of 28 against 7 of 60, $p = 1.00$). We therefore find no evidence that weaker designs cluster where low-income economies are sampled, although the comparison has little power, and the point estimate points the other way (low-income samples have more identification strategies). The test of whether study quality goes with a positive direction for output has almost none: only 3 of the 53 output studies were rated at moderate risk of bias, and all family-1 results are positive in 1 of those 3 against 11 of 50 other studies (Fisher $p = 0.54$).

Enabling conditions. Of the 170 heterogeneity entries, 135 are moderation tests. Human capital is tested in 15 studies (amplifies the effect in 9, dampens it in 3, mixed in 4), income group in 28 (31 entries: richer or higher-income group larger in 9, smaller in 7, mixed in 8, no difference in 5, unclear in 2), infrastructure or connectivity in 11 (5 amplify, 2 dampen, 2 mixed, 2 unclear), institutions or governance in 9 (3 amplify, 2 dampen, 5 mixed), trade openness in 8 (4 amplify, 2 dampen, 2 mixed), financial development in 6 and sector structure in 8 (6 mixed). A second reader agreed with the moderator type in 86% of a 28-entry sample drawn before the Google Scholar studies were added (kappa 0.83) but with the direction in only 71% (kappa 0.64), so the direction counts are indicative only. Amplification is the commonest finding for human capital and infrastructure, which is consistent with, but does not show, conditional returns (H2).

Thresholds. The studies state 27 thresholds or split points in 19 studies, for the human-capital index, digital maturity, internet penetration, trade openness, private credit and others. They are in different units, rest on one study each and cannot be pooled. Where two studies address the same variable they do not give comparable values.

Evidence gaps. Figure 6 shows that low-income economies are listed in 18 of 53 output studies, 4 of 15 labour-market studies, 2 of 13 distribution studies and 2 of 6 resource-cost

studies. High-income economies are listed in 31 of 53 output studies and 12 of 15 labour-market studies. The historical-wave and AI streams look alike on income-group comparison (42% against 41%) but differ on risk of bias (critical in 10% against 24%) and first sample year (median 2000 against 2012); the AI stream lists a low-income economy in 2 of 17 studies against 26 of 71 (Fisher $p = 0.08$).

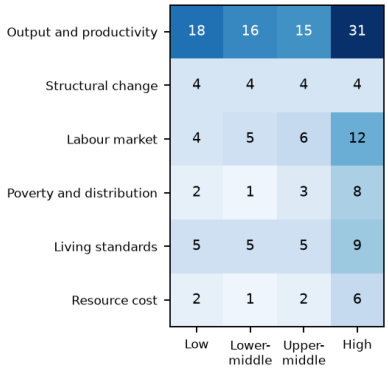


Figure 6: Evidence-gap map: studies per outcome family by income groups listed in the sample (a study counts in each group it lists; studies that do not state coverage are not shown).

5 Results of the macro-panel analysis

5.1 Mobile adoption and growth by income group (A1, A2)

Table 4 shows that a 10-subscription increase in mobile adoption per 100 people goes with higher annual growth in all four income groups over the same five-year period. The association is largest in lower-middle income economies (0.47 percentage points, 95% CI 0.16 to 0.77) and about 0.16 in upper-middle and high-income economies; the low-income estimate (0.36) is imprecise (CI −0.05 to 0.76, 20 economies). With country fixed effects, the lower-middle estimate remains (0.36, $p = 0.03$), and the others move toward 0.13 to 0.20. Extending the sample to 2023 gives the same pattern (lower-middle 0.34, $p < 0.01$; upper-middle 0.05, not significant; Appendix Table 21).

The predetermined specification, which uses the previous period's change, does not reproduce this pattern: the association is positive only in lower-middle income economies (0.26, $p = 0.02$), near zero in upper-middle, and negative in low-income (−0.38, $p = 0.13$) and high-income (−0.16, $p = 0.01$) economies, and negative in every income group for internet use (Appendix Table 19). The contemporaneous associations therefore partly reflect growth driving adoption or shared shocks, and they should not be read as a payoff to adoption. Results for fixed broadband rest on extrapolation from very few observations and are not interpreted (Appendix Table 20).

5.2 Dose-response and differences inside low-income economies (A3, A6)

Across baseline adoption levels, the association is largest where adoption was below 10 per 100 (0.41) and smaller at 40 to 80 (0.18); it is not monotone, since it is 0.27 above 80 per 100. For low and lower-middle income economies the point estimate above 80 per 100 is the largest (1.37) but

Table 4: Mobile adoption and annual GDP-per-capita growth by income group: change in annual growth (percentage points) per +10 subscriptions per 100 people, five-year periods 1995–2019 (95% CI; standard errors clustered by country).

Income group	Economies	Same period (A1)	Previous period (A2)
Low	20	0.36 [−0.05, 0.76]	−0.38 [−0.86, 0.11]
Lower-middle	42	0.47 [0.16, 0.77]	0.26 [0.03, 0.49]
Upper-middle	51	0.16 [0.04, 0.28]	0.01 [−0.15, 0.16]
High	64	0.16 [0.05, 0.26]	−0.16 [−0.28, −0.03]

imprecise (95% CI −0.01 to 2.74, 35 observations; Appendix Table 23). Inside low and lower-middle income economies, none of initial secondary enrolment, electricity access, private credit or initial income significantly changed the association (all $p > 0.19$; Appendix Table 24); the intervals are wide, and a standard-deviation change in a moderator could change the association by about 0.3 percentage points in either direction.

5.3 Event study around mobile takeoff (A4b)

The first event-study design, stacked cohorts with controls that adopt more than ten years later or never, was not interpretable (Section 3, Appendix H): nearly every economy crosses the threshold in the window, only 7 clean controls existed for the low and lower-middle income group, and that group showed a pre-trend of 12 log points. The replacement design compares each cohort with economies not yet treated at each horizon and adjusts for initial income and earlier growth. It passes the pre-trend rule in all four samples when the takeoff threshold is 10 subscriptions per 100 (mean placebo effect between −0.9 and 0.6 log points, all intervals including zero; Table 5).

Table 5: Event study around the year mobile subscriptions reach 10 per 100 people: effect on log GDP per capita (×100) relative to the year before takeoff and to economies not yet treated (95% bootstrap interval, 400 draws). Pre-trend: mean of the placebo effects at −5 to −2 years.

Sample	Treated	Pre-trend	Effect at +5 years	Effect at +10 years
All economies	178	−0.0 [−0.6, 0.4]	13.5 [0.6, 36.5]	47.8 [−12.5, 149.1]
Low and lower-middle	66	0.6 [−0.6, 1.6]	−5.9 [−18.6, 31.9]	not estimable
Upper-middle	56	−0.9 [−2.0, 0.3]	11.9 [−8.0, 46.7]	12.2 [−70.1, 118.6]
High	56	−0.4 [−1.2, 0.5]	19.1 [6.2, 36.7]	53.8 [4.2, 152.6]

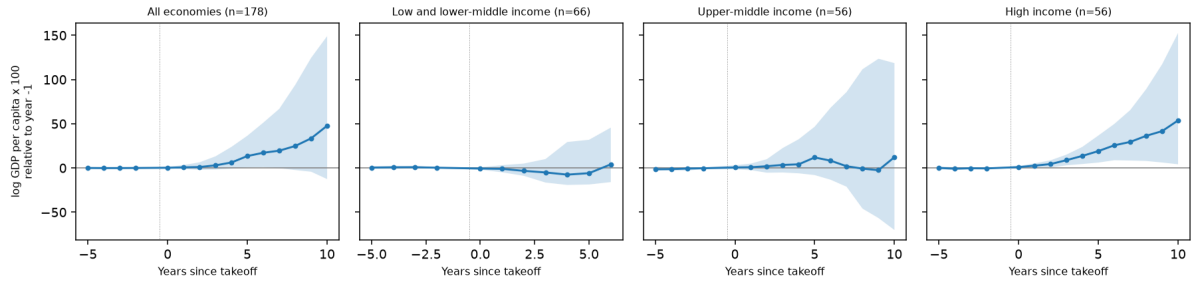


Figure 7: Event-study estimates by income group (takeoff at 10 mobile subscriptions per 100; shaded: 95% bootstrap interval). The low and lower-middle panel stops at six years because fewer than eight not-yet-treated economies exist at longer horizons.

Economies that reach 10 subscriptions per 100 earlier have higher GDP per capita five years later than comparable economies that reach it later, by about 14 log points (95% CI 0.6 to 36.5). The difference is distinguishable from zero only for high-income economies (19 log points, CI 6.2 to 36.7); the interval for low and lower-middle income economies, −18.6 to +31.9, is wide enough to contain effects of either sign and of the high-income size. Magnitudes at ten years (about 48 log points overall) are implausibly large for an effect of mobile telephony alone. They mainly show that economies reaching the threshold early are on different growth paths from those reaching it later, in ways initial income and earlier growth do not capture, and that at long horizons only the earliest cohorts contribute. The results are similar with a threshold of 25 per 100 (high income: 15.7 at five years, CI 6.5 to 28.8; low and lower-middle: −7.5, CI −17.6 to 13.3). With a threshold of 50, the pooled sample and the upper-middle sample fail the pre-trend rule and are not interpreted; the low and lower-middle estimate is 4.3 (CI −10.7 to 13.9) and the high-income estimate 8.9 (CI −0.8 to 20.8) (Appendix Tables 25 and 26).

Outcomes beyond income. For samples that pass the pre-trend rule, life expectancy is about 0.9 years higher five years after takeoff in the pooled sample (CI 0.1 to 1.7) and 0.8 years in high-income economies (CI 0.0 to 1.5), with a larger but imprecise estimate for low and lower-middle income economies (2.6 years, CI −0.5 to 6.4). Under-5 mortality is 15 log points lower in low and lower-middle income economies (CI −26.0 to 1.0). Unemployment and the agriculture share fail the pre-trend rule in most samples, and electricity use is not distinguishable from zero (Appendix Table 27). These outcomes have strong secular trends, so we read these as descriptive.

5.4 Local projections (A7)

Local projections give a dynamic profile without a binary treatment (Table 6, Figure 8). For lower-middle income economies, a 10-subscription increase over three years is followed by cumulative growth that is higher by 0.41 percentage points after one year and by 2.0 points after six years (CI 0.8 to 3.2), then 1.7 after eight. For upper-middle and high-income economies, the profile is flat, and the high-income point estimate turns slightly negative (cumulative −0.5 after eight years, CI −1.2 to 0.2). For low-income economies the profile starts positive and turns negative (−4.3 after eight years, CI −9.2 to 0.6), but the intervals include zero except as noted. The placebo (future adoption against past growth) is close to zero in every group, though with a wide interval for low-income economies (−0.75, CI −2.5 to 1.0).

Table 6: Local projections: cumulative growth of GDP per capita (percentage points) h years after a +10 change in mobile subscriptions per 100 over the previous three years (95% CI; country and year fixed effects).

Group	Economies	$h = 1$	$h = 2$	$h = 4$	$h = 8$
Low	24	0.25 [−0.08, 0.58]	0.34 [−0.35, 1.04]	−0.44 [−2.23, 1.35]	−4.31 [−9.19, 0.57]
Lower-middle	47	0.41 [0.11, 0.71]	0.76 [0.25, 1.27]	1.51 [0.69, 2.33]	1.69 [0.32, 3.07]
Upper-middle	58	−0.00 [−0.21, 0.21]	0.10 [−0.19, 0.38]	0.22 [−0.45, 0.88]	0.17 [−0.99, 1.34]
High	74	0.08 [−0.04, 0.19]	0.06 [−0.16, 0.29]	−0.06 [−0.48, 0.35]	−0.51 [−1.23, 0.20]

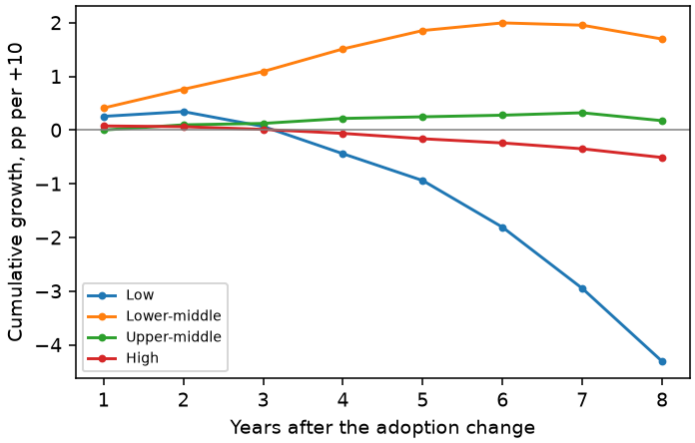


Figure 8: Local-projection profiles by income group.

5.5 Consistency across designs

Table 7 summarises which income group shows a positive association under each design. No group is positive under all four. Lower-middle income economies are positive under three (A1, A2, local projections); under the event study they are pooled with low-income economies and the pooled estimate is null with a negative point estimate. High-income economies are positive under the same-period regression and the event study, but flat or negative under the predetermined regression and the local projections. Upper-middle economies are positive only in the same-period regression. Low-income economies are never significantly positive. We conclude that the panel analysis does not support a statement that richer or poorer economies gain more; the answer depends on the design, which is the same message as in the reviewed studies.

Table 7: Direction and significance of the association between mobile adoption and GDP per capita growth, by design and income group (+ positive and significant at 5%, − negative and significant, 0 not significant; sign of the point estimate in parentheses).

Design	Low	Lower-middle	Upper-middle	High
A1 same period	0 (+)	+	+	+
A2 previous period	0 (−)	+	0 (+)	−
Event study (+5 years)	0 (−), low and lower-middle pooled		0 (+)	+
Local projections ($h = 6$)	0 (−)	+	0 (+)	0 (−)

6 Discussion

6.1 What the evidence supports

Four points stand out across the review and the panel analysis. First, the central estimate of the association between ICT-type adoption and output is positive but modest and very heterogeneous across studies (H1), and it shrinks when we allow for small-study bias. We find no support for H4: the GMM indicator changed sign when studies found later were added, and restricting to studies with an identification strategy raises the pooled value. A partial correlation of 0.16 means that adoption explains a few percent of the residual variation in output, and the

bias-corrected estimates are smaller still.

Second, returns look conditional on human capital, financial depth, governance and infrastructure (H2). Amplification is the commonest finding when authors test these conditions. This agrees with the long-run diffusion evidence that the intensity of use, rather than arrival, drives divergence [3], although the studies we synthesise cannot show that changing a moderator would change the return. The threshold results, such as a human-capital index below which the effect of the digital economy is negative in Africa, suggest that the conditions are not marginal but may bind. That reading needs replication, since each threshold comes from one study.

Third, the question of whether poorer or richer economies gain more cannot be answered from the reviewed evidence (H3). The comparisons that exist are few, mostly informal and often from weak designs, and low-income economies are rarely sampled. The measured association in the studies leans toward richer samples, but design differences could produce that lean.

Fourth, the timing of effects matters, but few studies examine it (H5). A study with a five-year lag found a positive effect where the contemporaneous effect was null. In our own analysis, predetermined and contemporaneous specifications give different pictures; the local projections show gains in lower-middle income economies building over about six years, and the event study shows gains in high-income economies emerging after the third year. Any claim about gains from a technology should state the horizon.

6.2 Reconciling the review with the panel analysis

The panel analysis adds a cautious observation. Under a single same-period design, the association between mobile adoption and growth is not smaller in poorer economies: it is largest in lower-middle income economies. This runs against the lean in the reviewed studies toward larger measured gains in richer samples. But the picture depends on the design: the event study finds a distinguishable gain only for high-income economies, and local projections only for lower-middle income economies, so the panel analysis does not settle which group gains more either. The two sources differ in technology (the panel covers mobile subscriptions only), in outcome (growth of GDP per capita) and in design (we use one specification for all groups, while the studies use different specifications by group), so they are not in direct conflict. The predetermined version does not reproduce the association, so we do not take it as evidence that poorer economies gain more. The honest summary is that the two sources do not agree on a difference by income, and neither identifies a causal effect.

6.3 AI

The evidence on AI is too thin and short to be put next to the evidence on earlier waves (H6). The few studies that exist show patterns similar to those for ICT where they test them: larger measured effects in richer economies and in high-skill occupations. But the exposure measures are indices of readiness or occupational exposure, the panels are short, 24% of the AI studies are rated critical, and only two list a low-income economy. A comparison with earlier technologies would require evidence on realised use, which none of the included studies has.

6.4 Implications

For policy, the safest reading is that digital investment is unlikely to pay off equally everywhere and that complementary capabilities matter. The review does not support statements of the form that a given income group should expect a given payoff. Three practical points follow. Evaluations of digital programmes should report effects at several horizons, because contemporaneous estimates can mislead. Programmes in low-income settings should be paired with, or at least monitored against, the complements the literature highlights, such as skills, electricity and finance. And claims about AI should be treated as provisional until studies with measures of realised use and credible designs appear.

6.5 Research agenda

For research, the main need is identification for developing and low-income economies, with instruments or timing designs of the kind used for submarine cables [6], and for AI, measures of realised use rather than readiness. Second, studies should report group-specific effects with a test of the difference, which most studies do not. Third, a registered, two-reviewer systematic review with Scopus, Web of Science and a complete Google Scholar search would reduce the selection concerns listed below. Fourth, the thresholds reported in single studies should be tested for replication across samples.

7 Limitations

- **Selection.** Only studies with a free or supplied full text were read: about 140 of the 247 records sent to full-text assessment are unread, and paywalled studies are over-represented among them. Records from four venues the author judged unreliable were skipped. Search covered OpenAlex, Crossref and NBER, plus the first page of four Google Scholar queries; Scopus, Web of Science and EconLit were not run, and eleven Google Scholar queries have not been run. The main search had poor recall (29 of 40 first-page Google Scholar results were missing from it), so the review is not exhaustive and its pooled results could shift as studies are added, as they did when a few were.
- **Single reviewer.** Screening used one assisted reviewer. A blind re-screening of a random sample by a separate model run gave moderate agreement at the abstract stage (79%, kappa 0.59) and suggests that the screening missed records a careful reader would include (Section 4.5). Extraction was by machine with a second machine pass and then checked on a random sample by the author by hand; the rest of the data has not been checked by a person, and no second human reviewer verified it independently.
- **Registration.** The protocol was not registered before the search. The certainty of the evidence was not graded (GRADE was not applied).
- **Design quality.** No study has low risk of bias; most treat adoption as exogenous. We report associations, not causal effects. First-pass risk-of-bias ratings are used; eleven ratings would change under the second pass, and the blind re-extraction agreed on the overall rating for only 11 of 16 studies, so the ratings are subjective.

- **Counting.** Direction counts are not weighted by size or quality and are not effect sizes. Study-level income comparisons mix World Bank income groups, regions and developed-developing splits.
- **Meta-regression.** Twenty-two papers with a headline estimate (24 with any usable estimate); the moderator estimates changed sign when a few studies were added, so they are unstable, and the clustered standard errors are optimistic. Partial correlations rely on an assumed degrees of freedom. Composite indices, levels and growth outcomes, and conditional main effects are mixed in one analysis despite exclusions. Two Asian studies (302 and 323) share authors, panel and index and are not independent. The sample is not a census, and the low- and middle-income group is small.
- **Language and period.** English only, 1995 onward.
- **Macro-panel analysis.** The World Bank panel analysis gives conditional associations only. Income groups use the current classification; the planned check with initial income was not run; CO2 emissions could not be retrieved. Roughly forty estimates were reported without correction for multiple testing.

8 Conclusion

Across 88 studies, ICT-type adoption is usually associated with higher output, but the association is modest, strongly heterogeneous, shrinks under a correction for small-study bias, is not lower in studies with credible identification, and is reported to be conditional on human capital, financial depth, governance and infrastructure. Whether rich or poor economies gain more is not established, and the evidence on AI is too thin to compare with earlier waves. A World Bank panel analysis finds that mobile adoption goes with faster growth in every income group in a same-period regression, most strongly in lower-middle income economies, but the pattern changes with the design (a staggered event study finds a gain only for high-income economies, and local projections only for lower-middle income economies), so without a causal design it is not a payoff. The search missed many relevant studies and adding some of them changed the moderator results, so those results should be treated as provisional. Credible identification for low- and middle-income economies and realised-use measures for AI are the main gaps.

Declarations

Data and code availability. The protocol, search and screening logs, extraction database, second-pass verification files, meta-regression inputs, World Bank panel and all code are in the project repository: `https://github.com/vinhqdang/Journal-of-Economy-and-Technology`. Full-text PDFs are not shared for copyright reasons.
Ethics. Not applicable; the review uses published studies and public statistics only.
Author contributions (CRediT). The review has a single author, Q.-V. Dang, who is responsible for all roles: conceptualisation, methodology, investigation, formal analysis, data curation, writing, supervision and project administration. The use of AI tools is described below.
Conflict of interest. The author declares no competing interests.
Funding. The author received no external funding for this work.
Use of AI tools. Generative AI tools were used to support this work: to polish the writing, to assist

with screening records and extracting data from the included studies (described in the Methods), and to help write and run analysis code. The ideas, the research question and direction, and the choice of methods and algorithms are the author's own. The author reviewed the AI-assisted output and takes full responsibility for the content of this paper.

Registration. The protocol was not registered before the search (see Methods).

References

[1] L.-H. Röller and L. Waverman (2001). Telecommunications infrastructure and economic development: a simultaneous approach. *American Economic Review* 91(4), 909–923. doi:10.1257/aer.91.4.909.

[2] D. Comin and B. Hobijn (2010). An exploration of technology diffusion. *American Economic Review* 100(5), 2031–2059. doi:10.1257/aer.100.5.2031.

[3] D. Comin and M. Mestieri (2018). If technology has arrived everywhere, why has income diverged? *American Economic Journal: Macroeconomics* 10(3), 137–178. doi:10.1257/mac.20150175.

[4] R. Jensen (2007). The digital provide: information (technology), market performance, and welfare in the South Indian fisheries sector. *Quarterly Journal of Economics* 122(3), 879–924. doi:10.1162/qjec.122.3.879.

[5] J. C. Aker and I. M. Mbiti (2010). Mobile phones and economic development in Africa. *Journal of Economic Perspectives* 24(3), 207–232. doi:10.1257/jep.24.3.207.

[6] J. Hjort and J. Poulsen (2019). The arrival of fast internet and employment in Africa. *American Economic Review* 109(3), 1032–1079. doi:10.1257/aer.20161385.

[7] D. H. Autor, F. Levy and R. J. Murnane (2003). The skill content of recent technological change: an empirical exploration. *Quarterly Journal of Economics* 118(4), 1279–1333. doi:10.1162/003355303322552801.

[8] D. Acemoglu and P. Restrepo (2019). Automation and new tasks: how technology displaces and reinstates labor. *Journal of Economic Perspectives* 33(2), 3–30. doi:10.1257/jep.33.2.3.

[9] W. M. Cohen and D. A. Levinthal (1990). Absorptive capacity: a new perspective on learning and innovation. *Administrative Science Quarterly* 35(1), 128–152. doi:10.2307/2393553.

[10] J. Dedrick, V. Gurbaxani and K. L. Kraemer (2003). Information technology and economic performance. *ACM Computing Surveys* 35(1), 1–28. doi:10.1145/641865.641866.

[11] M. J. Page et al. (2021). The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 372, n71. doi:10.1136/bmj.n71.

[12] D. Moher et al. (2015). Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic Reviews* 4, 1. doi:10.1186/2046-4053-4-1.

[13] M. Campbell et al. (2020). Synthesis without meta-analysis (SWiM) in systematic reviews: reporting guideline. *BMJ* 368, l6890. doi:10.1136/bmj.l6890.

[14] J. A. C. Sterne et al. (2016). ROBINS-I: a tool for assessing risk of bias in non-randomised studies of interventions. *BMJ* 355, i4919. doi:10.1136/bmj.i4919.

[15] T. D. Stanley and H. Doucouliagos (2012). *Meta-Regression Analysis in Economics and Business.* Routledge. doi:10.4324/9780203111710.

[16] T. D. Stanley and H. Doucouliagos (2014). Meta-regression approximations to reduce publication selection bias. *Research Synthesis Methods* 5(1), 60–78. doi:10.1002/jrsm.1095.

[17] R. DerSimonian and N. Laird (1986). Meta-analysis in clinical trials. *Controlled Clinical Trials* 7(3), 177–188. doi:10.1016/0197-2456(86)90046-2.

[18] B. Callaway and P. H. C. Sant'Anna (2021). Difference-in-differences with multiple time periods. *Journal of Econometrics* 225(2), 200–230. doi:10.1016/j.jeconom.2020.12.001.

[19] L. Sun and S. Abraham (2021). Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *Journal of Econometrics* 225(2), 175–199. doi:10.1016/j.jeconom.2020.09.006.

[20] C. de Chaisemartin and X. D'Haultfœuille (2020). Two-way fixed effects estimators with heterogeneous treatment effects. *American Economic Review* 110(9), 2964–2996. doi:10.1257/aer.20181169.

[21] Ò. Jordà (2005). Estimation and inference of impulse responses by local projections. *American Economic Review* 95(1), 161–182. doi:10.1257/0002828053828518.

A Included studies

Table 8: Studies included in the synthesis (n = 88). Authors are not listed because the extraction database does not store them; the DOI identifies each record. n.r. = not reported. RoB = overall risk of bias in the first-pass rating (moderate, serious, critical); no study was rated low.

ID	Year	Title	Technology	Econ.	RoB	DOI
10	2020	Financial inclusion, information and communication technology diffusion, and economic growth: a panel data analysis	ICT, Mobile, Internet	41	serious	https://doi.org/10.1080/02681102.2020.1734770
29	2019	The Internet of Things and economic growth in a panel of countries	ICT, Mobile, Internet	82	serious	https://doi.org/10.1080/10438599.2019.1695941
45	2024	Financial inclusion through digitalization and economic growth in Asia-Pacific countries	ICT, Mobile, Internet	30	serious	https://doi.org/10.1016/j.irfa.2024.103596
57	2020	Nexus between energy consumption, information and communications technology, and economic growth: An enquiry into emerging Asian countries	ICT, Mobile, Internet	10	serious	https://doi.org/10.1002/pa.2172
68	2022	Enhancing human development in developing regions: Do ICT and transport infrastructure matter?	ICT, Mobile, Internet	79	serious	https://doi.org/10.1016/j.techfore.2022.121725
71	2022	The ICT, financial development, energy consumption and economic growth nexus in MENA countries: dynamic panel CS-ARDL evidence	ICT, Mobile, Internet	11	serious	https://doi.org/10.1080/00036846.2022.2096861
88	2005	Does the digital divide matter? The role of information and communication technology in cross-country level and growth estimates	ICT, Mobile, Internet	94	serious	https://doi.org/10.1080/1043859042000304043
108	2024	New technologies and jobs in Europe	AI	16	moderate	https://doi.org/10.1093/epolic/eiae058
126	2023	Job creation and destruction in the digital age: Assessing heterogeneous effects across European Union countries	ICT	27	moderate	https://doi.org/10.1016/j.econmod.2023.106405

ID	Year	Title	Technology	Econ.	RoB	DOI
138	2023	Empirical Analysis of Inclusive Growth, Information and Communication Technology Adoption, and Institutional Quality	ICT, Mobile, Internet	193	serious	https://doi.org/10.3390/economies11040124
140	2015	What are the drivers of total factor productivity in the European Union?	ICT	17	serious	https://doi.org/10.1080/10438599.2015.1067007
162	2024	Impact of digitization on carbon productivity: an empirical analysis of 136 countries	Mobile, Internet	136	serious	https://doi.org/10.1038/s41598-024-55848-2
174	2025	Artificial intelligence and global carbon inequality: Addressing the challenges and opportunities for SDG 10, SDG 12, and SDG 13	AI	67	serious	https://doi.org/10.1016/j.gsf.2025.102072
217	2021	Does digital financial inclusion moderate or exacerbate output volatility?	Mobile	40	serious	https://doi.org/10.1080/13504851.2021.1963400
239	2018	What drives labor market polarization in advanced countries? The role of China and technology	ICT	19	serious	https://doi.org/10.1093/icc/dty063
245	2023	The impact of the digital economy on green total factor productivity in Belt and Road countries: the mediating role of energy transition	ICT, Internet	40	serious	https://doi.org/10.3389/fenvs.2023.1213961
268	2023	Informality and aggregate labor productivity growth: Does ICT moderate the relationship?	ICT, Internet	118	serious	https://doi.org/10.1016/j.telpol.2023.102681
269	2024	Economic growth and the proliferation of ICT infrastructures: which causes the other?	ICT, Mobile, Internet	41	serious	https://doi.org/10.1080/1331677x.2024.2310100
276	2025	Does artificial intelligence help in improving human capital based educational development? Evidence from 29 countries	AI	29	serious	https://doi.org/10.1016/j.techsoc.2025.103004
278	2024	Towards ICT diffusion and trade liberalisation on inclusive growth in Sub-Saharan Africa	ICT, Mobile, Internet	48	serious	https://doi.org/10.1007/s10668-023-04355-x
285	2022	ICT and Economic Growth in EU: A macro Level Comparison of Estimated ICT Output Elasticities	ICT	27	moderate	https://doi.org/10.1080/1097198x.2022.2094182
302	2013	Does ICT Participate in Economic Convergence among Asian Countries: Evidence from Dynamic Panel Data Model	ICT, Mobile, Internet	24	serious	https://doi.org/10.12948/issn14531305/17.2.2013.01
320	2022	The Effect of Technology Adoption on Financial Inclusion: A Cross-country Panel Analysis between China and Nigeria	Mobile, Internet	20	serious	https://doi.org/10.24018/ejbmr.2022.7.2.1314
321	2024	Information and communication technology and labour productivity growth: a production-frontier approach	ICT	24	serious	https://doi.org/10.1007/s10479-024-05818-8
323	2014	Total Factor Productivity, Demographic Traits and ICT: Empirical Analysis for Asia	ICT, Mobile, Internet	24	serious	https://doi.org/10.12948/issn14531305/18.1.2014.01
341	2024	The impact of information and communication technologies on export diversification: Evidence from developing countries	ICT, Mobile, Internet	110	moderate	https://doi.org/10.1080/09638199.2024.2419406
358	2023	Digital Divide, Credit Financing and Financial Inclusion: Changing Patterns of Poverty, Employment and Income Inequalities	Mobile, Internet	148	serious	n.a.

ID	Year	Title	Technology	Econ.	RoB	DOI
392	2018	TECHNOLOGICAL SOURCES OF ECONOMIC GROWTH IN EUROPE AND THE U.S.	ICT	10	serious	https://doi.org/10.3846/20294913.2017.1280557
393	2013	The Determinants of Total Factor Productivity in the EU: Insights from Sectoral Data and Common Dynamic Processes	ICT	17	serious	n.a.
407	2015	The Role of ICT in the Productivity of Central and Eastern European Countries: Cross-country Comparison	ICT, Internet	21	serious	n.a.
408	2025	Effects of the digital economy on economic growth in Africa: the role of human capital	ICT, Mobile, Internet	41	serious	https://doi.org/10.1080/23322039.2025.2525485
422	2024	Icts And Labour Productivity Nexus In Developing Countries: Evidence From Panel Estimation Approach	ICT, Mobile, Internet	84	serious	https://doi.org/10.33736/ijbs.6868.2024
441	2024	Artificial Intelligence Readiness and Employment: A Global Panel Analysis	AI	78	critical	https://doi.org/10.24818/18423264/58.4.24.04
442	2025	From Digital Divide to Employment Equity: How Digitalization Affects Women's Involvement in the Workforce in OECD Countries	Mobile, Internet	22	critical	https://doi.org/10.17583/generos.15672
448	2025	Unraveling the digital technologies and banking inclusion nexus in Sub Saharan Africa	Mobile, Internet	45	serious	https://doi.org/10.1007/s43621-025-01694-9
535	2026	The Economic Value of Agentic AI: A Comparative Analysis of Its Impact on Growth and Business Productivity in Developed and Emerging Economies	AI	10	critical	https://doi.org/10.9734/ajrcos/2026/v19i5859
617	2003	Bridging the digital divide: the growth implications of e-commerce for small and developing states	Internet	205	critical	n.a.
707	2026	Digitalisation and structural change: Evidence from cross-country analysis	Mobile, Internet	51	serious	https://doi.org/10.1016/j.jdec.2026.03.003
750	2025	Artificial Intelligence and Labour Markets in Southeast Asia: An Empirical Examination	AI	1	serious	https://doi.org/10.46557/001c.132415
755	2026	Artificial intelligence investment and renewable energy for environmental sustainability in ASEAN-5 under the foreign direct investment threshold	AI	5	serious	https://doi.org/10.1007/s43621-026-03998-w
802	2015	Differences in Job De-Routinization in OECD Countries: Evidence from PIAAC	ICT	22	serious	n.a.
803	2020	The Race between Technological Progress and Female Advancement: Changes in Gender and Skill Premia in OECD Countries	ICT	11	moderate	n.a.
879	2026	AI capability and labor productivity in low- and middle-income countries: An absorptive capacity approach	AI, ICT, Mobile, Internet	63	serious	https://doi.org/10.1016/j.technovation.2026.103645
891	2026	Can Artificial Intelligence Drive Sustainable Growth? Empirical Evidence on the AI-Energy-Growth Nexus in Advanced Economies	AI, ICT	9	serious	https://doi.org/10.3390/su18052308
913	2026	Estimation of Firm Labour Productivity and Sales Growth from Artificial Intelligence in Sub-Saharan African Countries	AI	10	critical	https://doi.org/10.12688/f1000research.179912.1

ID	Year	Title	Technology	Econ.	RoB	DOI
952	2026	Twin Transition and Labor Informality in Asia-Pacific: Threshold Effects and Income-Group Heterogeneity	ICT, Internet	17	serious	https://doi.org/10.3390/economies14090398
957	2024	IMPACT OF ICT ON ECONOMIC GROWTH: CASE OF OECD AND MENA COUNTRIES	ICT, Mobile, Internet	32	serious	https://doi.org/10.59671/dga69
988	2023	IMPACT OF DIGITALISATION ON PRODUCTIVITY GROWTH IN EU MEMBER STATES	ICT	27	serious	https://doi.org/10.30525/2661-5169/2023-3-2
1013	2025	The Effect of Artificial Intelligence Investments on Economic Development: Panel Data Analysis	AI	8	serious	https://doi.org/10.17694/bajece.1802500
1025	2005	Testing the impact of ICT in developed countries during 1980-1995: distributional analysis in Solow's growth accounting framework	ICT	15	serious	n.a.
1026	2026	The nonlinear link between AI and energy poverty: Empirical evidence from global data	AI	64	serious	https://doi.org/10.1016/j.jclepro.2026.148590
1031	2025	Exploring the Moderating impact of ICT Infrastructure on Trade Openness-Sectoral Growth Nexus in Sub-Saharan Africa (SSA)	ICT, Mobile, Internet	26	serious	https://doi.org/10.58709/niujhu.v10i3.2267
1192	2026	The Digital Maturity Paradox: The Divergent Impact of Network Readiness and FDI Across Development Tiers	ICT	121	critical	https://doi.org/10.3390/economies14080320
1372	2024	The relationship between inequality and poverty in developing countries: mitigating role of virtual social network and internet access in schools	Internet	57	serious	https://doi.org/10.1108/ijse-09-2023-0695
1419	2017	Mobile banking usage, quality of growth, inequality and poverty in developing countries	Mobile	93	serious	https://doi.org/10.1177/0266666917744006
1428	2021	IMPACT OF MOBILE PHONES AND INTERNET USE ON FINANCIAL INCLUSION: EMPIRICAL EVIDENCE FROM THE EU POST-COMMUNIST COUNTRIES	Mobile, Internet	11	serious	https://doi.org/10.3846/tede.2021.14508
1438	2021	Impact of Economic Integration and Information and Communication Technology on Economic Growth for European Union: Dynamic Panel GMM Approach	ICT, Mobile, Internet	27	serious	https://doi.org/10.5296/ber.v11i4.19064
1448	2023	The effect of information and communication technology on economic growth high-income countries	ICT, Mobile, Internet	10	serious	https://doi.org/10.55493/5002.v13i9.4824
1469	2026	Reducing Gender Inequality in Education in Developing Countries: Do Information and Communication Technologies Matter?	ICT, Mobile, Internet	64	serious	https://doi.org/10.1002/sd.71201
1494	2026	Digital transformation and income disparities: A threshold analysis in developing countries	ICT, Mobile, Internet	45	serious	https://doi.org/10.1556/032.2025.00183
1526	2022	Artificial Intelligence and Employment: New Cross-Country Evidence	AI	23	moderate	https://doi.org/10.3389/frai.2022.832736
1564	2025	The Impact of Artificial Intelligence on Employment: A Panel Data Analysis for Selected Countries	AI	29	serious	https://doi.org/10.30784/epfad.1621455
1580	2023	Implications of AI innovation on economic growth: a panel data study	AI	122	serious	https://doi.org/10.1186/s40008-023-00307-w

ID	Year	Title	Technology	Econ.	RoB	DOI
1596	2026	Impact of Generative AI Adoption on Micro and Small Enterprises in Emerging Economies	Other	3	critical	https://doi.org/10.64060/ijpssdv2i12
1645	2025	Cryptocurrency Adoption Patterns and Financial Inclusion Outcomes in Emerging Market Economies	Mobile, Internet, Crypto	15	critical	https://doi.org/10.71086/iajir/v12i4/iajir1244
1652	2025	Crypto-driven growth: A comparative study of Bitcoin and Ethereum on economic growth for multi-country analysis	Crypto	14	critical	https://doi.org/10.32609/j.ruje.11.164511
1670	2019	FINANCIAL INCLUSION, MOBILE PHONE DIFFUSION, AND ECONOMIC GROWTH; EVIDENCE FROM AFRICA	Mobile	32	serious	https://doi.org/10.32479/ijefi.8256
1679	2022	The Employment Effects of Mobile Internet in Developing Countries	Mobile, Internet	14	moderate	n.a.
1713	2012	ICT capital and labour productivity growth: A non-parametric analysis of 14 OECD countries	ICT	14	serious	https://doi.org/10.1016/j.telpol.2011.12.012
1759	2022	ICT capital–skill complementarity and wage inequality: Evidence from OECD countries	ICT	14	moderate	https://doi.org/10.1016/j.labeco.2022.102151
1798	2022	The role of governance in the effect of the internet on financial inclusion in sub-Saharan Africa	Internet	42	serious	https://doi.org/10.2139/ssrn.4315622
9001	2019	The association between mobile phones and the risk of brain cancer mortality: a 25-year cross-country analysis	Mobile	88	serious	https://doi.org/10.1111/coep.12456
9101	2026	ICT Infrastructure, Economic Growth and ASEAN Income Heterogeneity: A Panel Analysis	ICT, Mobile, Internet	10	serious	https://doi.org/10.1111/apel.70056
9102	2026	Internet and Economic Growth Across Income Levels: a Panel Data Analysis	Internet	78	serious	https://doi.org/10.1007/s13132-026-03136-x
9103	2022	ICT Diffusion, Industrialisation and Economic Growth Nexus: an International Cross-country Analysis	ICT, Mobile, Internet	171	serious	https://doi.org/10.1007/s13132-021-00795-w
9104	2024	Information and Communication Technology (ICT), Growth and Development in Developing Regions: Evidence from a Comparative Analysis and a New Approach	ICT, Mobile, Internet	73	serious	https://doi.org/10.1007/s13132-023-01571-8
9105	2021	The impact of ICT on economic growth-Comparing rich and poor countries	ICT, Mobile, Internet	123	serious	https://doi.org/10.1016/j.telpol.2020.102082
9106	2023	Mobile Phones and Local Economic Development: A Global Evidence	Mobile, Internet	120	moderate	n.a.
9201	2023	Does digitalization contribute to lesser income inequality? Evidence from G20 countries	Mobile, Internet	19	serious	https://doi.org/10.1080/02681102.2022.2123443
9203	2024	Impact of ICT Infrastructure, Financial Development, and Trade Openness on Economic Growth: New Evidence from Low- and High-Income Countries	ICT, Internet	85	serious	https://doi.org/10.1007/s13132-023-01332-7
9204	2021	The Effects of Information and Communication Technology (ICT) Use on Human Development: A Macroeconomic Approach	ICT, Mobile, Internet	130	serious	https://doi.org/10.3390/economies9030128
9302	2023	The effect of the digital economy on total factor productivity in European regions	ICT, Internet	27	serious	https://doi.org/10.1016/j.telpol.2023.102650

ID	Year	Title	Technology	Econ.	RoB	DOI
9303	2008	An exploration of the effects of the interaction between ICT and labor force on economic growth in transition economies	ICT	18	critical	https://doi.org/10.1016/j.ijpe.2008.06.008
9304	2020	Enhancing ICT for productivity in sub-Saharan Africa: Thresholds for complementary policies	Mobile, Internet	25	serious	n.a.
9305	2013	Information Technology and Productivity in Developed and Developing Countries	ICT	45	moderate	n.a.
9401	2024	Exploring the growth effects of Artificial Intelligence in developing countries in Africa using a semi-endogenous growth model	AI, ICT	48	critical	n.a.
9402	2022	Digital transformation, development and productivity in developing countries: is artificial intelligence a curse or a blessing?	ICT	25	serious	n.a.
9404	2025	Artificial intelligence (AI)-poverty-economic growth nexus in selected BRICS-Plus countries: does the moderating role of governance matter?	AI	5	serious	n.a.

B Results by outcome family

Table 9: Studies reporting output and productivity outcomes (n = 53). "Directions" counts the study's results in this family: + positive, − negative, 0 null, m mixed. Technology codes as in Table 8.

ID	Tech.	Econ.	Years	Estimator	Directions	RoB
10	ICT, Mobile, Internet	41	2004–2015	GMM	20, 5m	serious
29	ICT, Mobile, Internet	82	2010–2017	FE/RE/OLS	7+, 10	serious
45	ICT, Mobile, Internet	30	2014–2021	FE/RE/OLS	5+	serious
57	ICT, Mobile, Internet	10	2000–2017	ARDL/coint./heterog. panel	2+, 1−, 20, 1m	serious
71	ICT, Mobile, Internet	11	1980–2018	ARDL/coint./heterog. panel	3+	serious
88	ICT, Mobile, Internet	94	1983–1997	FE/RE/OLS	4+, 1m	serious
140	ICT	17	1995–2007	ARDL/coint./heterog. panel	1+	serious
217	Mobile	40	2009–2017	GMM	5+, 1m	serious
245	ICT, Internet	40	2006–2021	ARDL/coint./heterog. panel	3−, 2m	serious
268	ICT, Internet	118	1997–2017	FE/RE/OLS	3+, 3−, 30, 2m	serious
269	ICT, Mobile, Internet	41	2000–2019	ARDL/coint./heterog. panel	2+, 10	serious
285	ICT	27	1996–2016	FE/RE/OLS	6+	moderate
302	ICT, Mobile, Internet	24	2000–2010	GMM	2+, 1−	serious
321	ICT	24	1995–2019	Accounting/DEA/other	3+, 1−, 10, 2m	serious
323	ICT, Mobile, Internet	24	2000–2010	GMM	2+	serious
358	Mobile, Internet	148	2008–2021	FE/RE/OLS	1+	serious
392	ICT	10	1980–2010	Accounting/DEA/other	3+, 10	serious
393	ICT	17	1995–2007	ARDL/coint./heterog. panel	3+, 10	serious
407	ICT, Internet	21	1993–2011	FE/RE/OLS	4+, 10, 1m	serious
408	ICT, Mobile, Internet	41	2000–2021	GMM	5+, 1−	serious
422	ICT, Mobile, Internet	84	2000–2019	GMM	5+, 1m	serious
535	AI	10	2015–2024	FE/RE/OLS	7+	critical
617	Internet	205	2000	Accounting/DEA/other	2+	critical
879	AI, ICT, Mobile, Internet	63	2005–2019	GMM	7+, 1m	serious
891	AI, ICT	9	2010–2025	ARDL/coint./heterog. panel	1+, 1−, 20	serious
913	AI	10	2007–2024	FE/RE/OLS	4+, 20, 2m	critical
952	ICT, Internet	17	2005–2022	GMM	2+	serious
957	ICT, Mobile, Internet	32	2010–2023	GMM	3+, 40, 1m	serious
988	ICT	27	2017–2022	FE/RE/OLS	2+, 10	serious

ID	Tech.	Econ.	Years	Estimator	Directions	RoB
1025	ICT	15	1980–1995	Accounting/DEA/other	20, 2m	serious
1192	ICT	121	2023	FE/RE/OLS	1+, 1−, 10, 1m	critical
1419	Mobile	93	2005–2011	FE/RE/OLS	2m	serious
1438	ICT, Mobile, Internet	27	2002–2019	GMM	2+, 2m	serious
1448	ICT, Mobile, Internet	10	2001–2020	FE/RE/OLS	40	serious
1580	AI	122	1970–2019	GMM	5+, 10, 1m	serious
1596	Other	3	2025	FE/RE/OLS	2+	critical
1652	Crypto	14	2015–2023	ARDL/coint./heterog. panel	3+, 2m	critical
1670	Mobile	32	2004–2016	Accounting/DEA/other	1+, 1−, 20, 1m	serious
1713	ICT	14	1995–2005	Accounting/DEA/other	1+, 2−, 10, 1m	serious
9101	ICT, Mobile, Internet	10	1995–2020	FE/RE/OLS	5+, 1−, 3m	serious
9102	Internet	78	1999–2021	FE/RE/OLS	2−, 20, 1m	serious
9103	ICT, Mobile, Internet	171	2000–2018	GMM	6+	serious
9104	ICT, Mobile, Internet	73	2000–2018	GMM	2+, 3−	serious
9105	ICT, Mobile, Internet	123	2002–2017	GMM	5+	serious
9106	Mobile, Internet	120	1998–2018	Accounting/DEA/other	5+, 1m	moderate
9203	ICT, Internet	85	2000–2019	GMM	5+, 5m	serious
9302	ICT, Internet	27	2006–2020	GMM	3+, 2m	serious
9303	ICT	18	1993–2002	FE/RE/OLS	1+, 1−, 20	critical
9304	Mobile, Internet	25	1980–2014	GMM	2+, 10, 1m	serious
9305	ICT	45	1994–2007	FE/RE/OLS	4+, 10	moderate
9401	AI, ICT	48	2007–2020	GMM	6+, 1−, 10	critical
9402	ICT	25	2017–2017	FE/RE/OLS	2+	serious
9404	AI	5	2012–2023	ARDL/coint./heterog. panel	2+, 1m	serious

Table 10: Studies reporting structural change outcomes (n = 6). "Directions" counts the study's results in this family: + positive, − negative, 0 null, m mixed. Technology codes as in Table 8.

ID	Tech.	Econ.	Years	Estimator	Directions	RoB
239	ICT	19	1996–2010	FE/RE/OLS	20	serious
341	ICT, Mobile, Internet	110	1995–2019	GMM	6−	moderate
707	Mobile, Internet	51	1990–2018	GMM	4+, 3−	serious
1031	ICT, Mobile, Internet	26	1991–2023	ARDL/coint./heterog. panel	3+, 5−, 20	serious
1679	Mobile, Internet	14	2000–2017	FE/RE/OLS	2+, 10	moderate
9103	ICT, Mobile, Internet	171	2000–2018	GMM	5+	serious

Table 11: Studies reporting labour market outcomes (n = 15). "Directions" counts the study's results in this family: + positive, − negative, 0 null, m mixed. Technology codes as in Table 8.

ID	Tech.	Econ.	Years	Estimator	Directions	RoB
108	AI	16	2011–2019	FE/RE/OLS	2+, 1−, 30	moderate
126	ICT	27	1995–2019	GMM	4+, 1m	moderate
239	ICT	19	1996–2010	FE/RE/OLS	3+, 10, 1m	serious
358	Mobile, Internet	148	2008–2021	FE/RE/OLS	1+, 1−, 10	serious
441	AI	78	2019–2022	FE/RE/OLS	3+, 20, 1m	critical
442	Mobile, Internet	22	2000–2022	ARDL/coint./heterog. panel	2+, 1m	critical
750	AI	1	2014–2020	FE/RE/OLS	2−, 1m	serious
802	ICT	22	2011–2012	FE/RE/OLS	1+, 3−	serious
803	ICT	11	1980–2008	GMM	2+, 2−, 1m	moderate
952	ICT, Internet	17	2005–2022	GMM	2+, 1−, 40, 1m	serious
1526	AI	23	2010–2019	FE/RE/OLS	2+, 2−, 20, 1m	moderate
1564	AI	29	2017–2021	GMM	2+, 1−	serious
1679	Mobile, Internet	14	2000–2017	FE/RE/OLS	6+, 1−, 10, 1m	moderate
1759	ICT	14	1970–2005	GMM	1+	moderate
9402	ICT	25	2017–2017	FE/RE/OLS	2+, 20	serious

Table 12: Studies reporting poverty and distribution outcomes (n = 13). "Directions" counts the study's results in this family: + positive, − negative, 0 null, m mixed. Technology codes as in Table 8.

ID	Tech.	Econ.	Years	Estimator	Directions	RoB
162	Mobile, Internet	136	2000–2020	GMM	1−	serious
174	AI	67	1995–2019	GMM	8+, 1m	serious
276	AI	29	2017–2021	FE/RE/OLS	1+, 1m	serious
278	ICT, Mobile, Internet	48	2005–2020	GMM	5+	serious
358	Mobile, Internet	148	2008–2021	FE/RE/OLS	1+, 3−	serious
802	ICT	22	2011–2012	FE/RE/OLS	10	serious
803	ICT	11	1980–2008	GMM	1+, 1−, 10	moderate
1372	Internet	57	2012–2016	FE/RE/OLS	1+, 2−	serious
1419	Mobile	93	2005–2011	FE/RE/OLS	20, 2m	serious
1494	ICT, Mobile, Internet	45	2000–2023	GMM	1+, 1−, 10, 2m	serious
1759	ICT	14	1970–2005	GMM	3+, 1−, 1m	moderate
9201	Mobile, Internet	19	2002–2018	GMM	2+, 6−, 20, 1m	serious
9404	AI	5	2012–2023	ARDL/coint./heterog. panel	1+, 1−, 1m	serious

Table 13: Studies reporting living standards beyond income outcomes (n = 16). "Directions" counts the study's results in this family: + positive, − negative, 0 null, m mixed. Technology codes as in Table 8.

ID	Tech.	Econ.	Years	Estimator	Directions	RoB
10	ICT, Mobile, Internet	41	2004–2015	GMM	1m	serious
68	ICT, Mobile, Internet	79	1990–2018	GMM	6+, 10, 1m	serious
138	ICT, Mobile, Internet	193	2010–2019	GMM	4+, 1−, 4m	serious
276	AI	29	2017–2021	FE/RE/OLS	7+, 1−, 10, 2m	serious
320	Mobile, Internet	20	2005–2020	FE/RE/OLS	3+, 20, 1m	serious
358	Mobile, Internet	148	2008–2021	FE/RE/OLS	1+	serious
448	Mobile, Internet	45	2000–2023	FE/RE/OLS	5+, 10, 1m	serious
1013	AI	8	2012–2023	ARDL/coint./heterog. panel	2+, 10	serious
1026	AI	64	2000–2019	GMM	1−, 2m	serious
1428	Mobile, Internet	11	1996–2017	ARDL/coint./heterog. panel	3+, 20, 1m	serious
1469	ICT, Mobile, Internet	64	2001–2022	GMM	7−, 1m	serious
1645	Mobile, Internet, Crypto	15	2018–2023	Accounting/DEA/other	1+	critical
1798	Internet	42	2004–2020	GMM	1+, 2−	serious
9001	Mobile	88	1960–2015	FE/RE/OLS	4+, 30, 1m	serious
9104	ICT, Mobile, Internet	73	2000–2018	GMM	3+, 1−	serious
9204	ICT, Mobile, Internet	130	2007–2019	GMM	2+, 20	serious

Table 14: Studies reporting resource cost outcomes (n = 6). "Directions" counts the study's results in this family: + positive, − negative, 0 null, m mixed. Technology codes as in Table 8.

ID	Tech.	Econ.	Years	Estimator	Directions	RoB
162	Mobile, Internet	136	2000–2020	GMM	4+, 1m	serious
174	AI	67	1995–2019	GMM	1+	serious
245	ICT, Internet	40	2006–2021	ARDL/coint./heterog. panel	1+	serious
755	AI	5	2015–2023	FE/RE/OLS	4−, 50	serious
891	AI, ICT	9	2010–2025	ARDL/coint./heterog. panel	10, 1m	serious
1026	AI	64	2000–2019	GMM	1−, 1m	serious

C Search strategy

Search blocks (protocol)

Four concept blocks were combined with AND and searched in title and abstract. The OpenAlex query below is the exact string that was run (restricted to the economics, econometrics and

finance field), and the Crossref and NBER searches used 96 combinations of a technology term and a context term.

- **Block A (technology):** information and communication technolog*, ICT, internet, broadband, mobile phone*, mobile telephon*, smartphone*, digital technolog*, digital economy, artificial intelligence, machine learning, deep learning, generative AI, large language model*, general purpose technolog*, cryptocurrenc*, bitcoin, crypto-asset*, blockchain.
- **Block B (economy level):** countr*, cross-country, nation*, developing econom*, emerging econom*, low-income, middle-income, high-income, income group*, developed econom*, advanced econom*.
- **Block C (outcomes):** growth, productivity, GDP, poverty, inequality, employment, wage*, labor or labour share, structural transformation, financial inclusion, welfare, living standard*, human development.
- **Block D (heterogeneity):** heterogene*, absorptive capacity, complementar*, threshold*, human capital, digital divide, convergence, divergence, technology diffusion, technology adoption, leapfrog*.

OpenAlex query as run (2 October 2026)

```
title_and_abstract.search.exact:("information and communication technology" OR "information and communication technologies" OR ICT OR internet OR broadband OR "mobile phone" OR "mobile phones" OR "mobile telephony" OR "mobile telephone" OR smartphone* OR "digital technology" OR "digital technologies" OR "digital economy" OR "artificial intelligence" OR "machine learning" OR "deep learning" OR "generative AI" OR "large language model" OR "large language models" OR "general purpose technology" OR "general-purpose technology" OR "general purpose technologies" OR cryptocurrenc* OR bitcoin OR "crypto-asset" OR "crypto-assets" OR blockchain) AND (countr* OR "cross-country" OR "cross country" OR nation* OR "developing economy" OR "developing economies" OR "emerging economy" OR "emerging economies" OR low-income OR middle-income OR high-income OR "income group" OR "income groups" OR "developed economy" OR "developed economies" OR "advanced economy" OR "advanced economies") AND (growth OR productivity OR GDP OR poverty OR inequality OR employment OR wage* OR "labor share" OR "labour share" OR "structural transformation" OR "financial inclusion" OR welfare OR "living standard" OR "living standards" OR "human development") AND (heterogene* OR "absorptive capacity" OR complementar* OR threshold* OR "human capital" OR "digital divide" OR convergence OR divergence OR "technology diffusion" OR "technology adoption" OR
leapfrog*),from_publication_date:1995-01-01,type:article|preprint,language:en,primary_topic.field.id:20
```

Crossref and NBER searches

Technology terms: internet; broadband; mobile phones; information and communication technology; digital technology; artificial intelligence; generative AI; machine learning; cryptocurrency; bitcoin; technology diffusion; general purpose technology. Context terms: cross-country panel economic growth; developing countries income groups; digital divide; poverty and inequality countries; productivity emerging economies; employment and wages across countries; absorptive capacity human capital; financial inclusion. All pairs (12 by 8 = 96) were run from 1995.

Google Scholar queries run (4 October 2026, first page of 10 results each)

- G1: (“mobile phones” OR “mobile telephony” OR internet OR broadband OR ICT) “economic growth” “developing countries” “income groups” panel cross-country
- G2: “information and communication technology” growth “low-income” “middle-income” “high-income” countries panel GMM
- G4: (internet OR mobile OR ICT) “labor productivity” OR “total factor productivity” countries “human capital” OR institutions interaction OR threshold
- G7: “artificial intelligence” “economic growth” OR productivity countries panel “developing countries” OR “emerging economies” OR “income”

Records by source

Source	Distinct records
OpenAlex	1,840
Crossref	1,285
NBER working papers	579
arXiv	0 (attempted, not completed)
Scopus, Web of Science, EconLit	0 (not run)
Google Scholar (queries G1, G2, G4, G7, first page each)	40 results, 29 not in the main search

D Eligibility, extraction form and risk-of-bias criteria

Eligibility

Criterion	Include	Exclude
Study design	Quantitative empirical studies and meta-analyses with their own estimation strategy	Opinion pieces, purely conceptual papers, case descriptions without estimation, simulation-only studies
Sample, historical waves	At least 10 economies, or an explicit comparison of income groups	Single-country and few-country studies
Sample, AI	At least 3 economies, or an explicit comparison across income levels	Single-country firm or worker studies are listed separately as supplementary evidence
Exposure	ICT in general, mobile, internet or broadband, smartphones and mobile data, crypto-assets, AI	Technology used as a control only
Outcome	At least one of six families: output and productivity; structural change; labour market; poverty and distribution; living standards beyond income; resource cost	Adoption or diffusion as the only outcome (kept as context)
Period and language	1995 to the search date; English	Earlier; other languages
Status	Peer-reviewed articles and working papers from named series; preprints flagged	Conference abstracts, blog posts, reports without methods

Exclusion codes at full text: E1 too few economies and no income-group comparison; E2 exposure not eligible; E3 no outcome in the six families; E4 conceptual, review, or no own estimation; DUP duplicate or earlier version of another paper.

Extraction form (frozen before extraction)

For each study: decision and coded reason; stream (historical waves, AI or both); technology and exposure measure; sample (number of economies, groups, income coverage, years, unit); design (estimator, identification strategy, controls); results by outcome family (indicator, direction,

estimate, uncertainty, horizon, locator and quote); heterogeneity findings (moderator, finding, threshold, locator); results by income group; number of specifications; whether the headline estimate is the preferred specification; risk-of-bias ratings with reasons; venue concern; free-text notes. Every extracted result carries a page or table locator and a verbatim quote of at most 25 words.

Risk-of-bias domains and rating guide

1. Confounding and reverse causation: is adoption treated as exogenous, and is there a credible instrument or timing design?
2. Exposure measurement: does the adoption measure capture use, or only access or investment?
3. Selection of countries and years: missing-data patterns and survivorship.
4. Outcome measurement: quality of the national statistics used.
5. Specification and researcher degrees of freedom: number of specifications and sensitivity.
6. Selective reporting.

Rating guide: *low* = credible design for the claim; *moderate* = standard panel design with reasonable controls but exogeneity not established; *serious* = causal language with no strategy for reverse causation, or a measure that does not capture the exposure; *critical* = the conclusions cannot be trusted from this design.

E Meta-regression inputs

Table 15: Estimates used in the meta-regression (148 estimates from 24 studies): partial correlation (PCC) and its standard error, computed from the printed coefficient and uncertainty with degrees of freedom equal to observations minus 10. The second half of the list continues in the right-hand block; t-statistics are available with the data and code.

Estimate	Role	Obs.	PCC	SE	Estimate	Role	Obs.	PCC	SE
10-1	headline	395	0.284	0.049	393-1	headline	1106	0.060	0.030
10-2	baseline	395	-0.355	0.048	393-2	baseline	1322	0.038	0.028
10-3	headline	392	0.167	0.050	393-3	robustness	1322	0.082	0.028
10-4	baseline	392	-0.220	0.050	393-4	robustness	1158	0.070	0.029
10-5	baseline	397	-0.008	0.051	393-5	robustness	1158	0.044	0.029
10-6	baseline	396	-0.018	0.051	407-1	headline	399	0.544	0.043
29-1	headline	574	0.107	0.042	407-10	robustness	98	0.009	0.107
29-10	robustness	450	0.275	0.046	407-11	robustness	112	-0.194	0.097
29-11	robustness	252	0.210	0.063	407-12	robustness	56	-0.218	0.144
29-12	robustness	378	0.147	0.052	407-13	robustness	56	0.405	0.135
29-13	robustness	574	0.093	0.042	407-2	subsample	133	0.560	0.075
29-14	robustness	574	0.093	0.042	407-3	subsample	266	0.138	0.062
29-2	baseline	574	0.109	0.042	407-5	subsample	133	0.411	0.082
29-3	robustness	410	0.119	0.050	407-7	robustness	49	0.130	0.159
29-4	robustness	410	0.110	0.050	407-8	robustness	56	0.600	0.118
29-5	subsample	252	0.122	0.064	407-9	robustness	28	0.028	0.236
29-6	subsample	252	0.109	0.064	408-7	robustness	779	-0.072	0.036
29-7	subsample	322	0.108	0.056	617-1	headline	126	0.511	0.080
29-8	subsample	322	0.108	0.056	617-2	robustness	126	0.497	0.081
29-9	robustness	630	0.197	0.039	952-1	headline	198	0.123	0.072
45-1	headline	90	0.488	0.098	988-1	headline	168	0.000	0.080

Estimate	Role	Obs.	PCC	SE	Estimate	Role	Obs.	PCC	SE
45-2	subsample	90	0.616	0.088	988-2	robustness	168	0.000	0.080
45-3	subsample	90	0.408	0.102	988-3	robustness	168	0.000	0.080
45-4	robustness	90	0.488	0.098	988-4	robustness	168	0.000	0.080
45-5	robustness	90	0.327	0.106	988-6	robustness	140	0.000	0.088
71-1	headline	429	0.081	0.049	988-7	robustness	140	0.000	0.088
71-2	baseline	429	0.089	0.049	988-8	robustness	168	0.000	0.080
71-3	robustness	429	0.171	0.048	1438-1	headline	138	0.191	0.087
88-10	robustness	181	0.710	0.054	1438-2	baseline	138	0.157	0.087
88-7	robustness	465	0.100	0.047	9102-1	headline	780	-0.086	0.036
88-8	robustness	293	0.647	0.045	9102-10	robustness	346	-0.091	0.054
88-9	robustness	465	0.090	0.047	9102-2	subsample	99	0.070	0.106
140-1	headline	451	0.150	0.047	9102-3	subsample	335	-0.074	0.055
140-2	baseline	451	0.114	0.047	9102-4	subsample	346	-0.104	0.054
245-1	headline	628	-0.076	0.040	9102-5	baseline	780	-0.107	0.036
245-10	subsample	373	-0.183	0.052	9102-6	robustness	780	-0.107	0.036
245-11	subsample	239	-0.282	0.063	9102-7	robustness	780	-0.072	0.036
245-12	robustness	528	-0.245	0.043	9102-8	robustness	99	-0.079	0.106
245-13	robustness	605	-0.083	0.041	9102-9	robustness	335	-0.055	0.055
245-14	robustness	587	-0.239	0.040	9105-1	headline	628	0.117	0.040
245-2	baseline	628	-0.079	0.040	9105-10	robustness	572	0.109	0.042
245-3	robustness	587	-0.169	0.041	9105-11	robustness	587	0.486	0.036
245-4	robustness	605	-0.296	0.039	9105-12	robustness	130	0.301	0.087
245-5	robustness	604	-0.244	0.040	9105-2	headline	753	0.558	0.030
245-6	subsample	306	-0.186	0.057	9105-3	headline	224	0.360	0.064
245-7	subsample	94	0.168	0.108	9105-4	subsample	630	0.115	0.040
245-8	subsample	205	-0.158	0.071	9105-5	subsample	630	0.401	0.037
245-9	subsample	16	-0.561	0.338	9105-6	subsample	812	0.437	0.032
268-1	headline	2374	0.049	0.021	9105-7	subsample	812	0.549	0.030
268-10	subsample	791	0.078	0.036	9105-8	subsample	280	0.267	0.059
268-11	subsample	791	0.075	0.036	9105-9	subsample	275	0.400	0.056
268-12	subsample	791	0.093	0.036	9203-1	headline	513	0.045	0.045
268-2	robustness	2374	0.066	0.021	9203-10	baseline	1102	0.010	0.030
268-3	robustness	2374	0.045	0.021	9203-2	subsample	513	-0.024	0.045
268-4	robustness	2374	0.060	0.021	9203-3	subsample	513	0.008	0.045
268-5	subsample	1583	0.044	0.025	9203-4	subsample	513	0.025	0.045
268-6	subsample	1583	0.062	0.025	9203-5	headline	1160	0.025	0.029
268-7	subsample	1583	0.041	0.025	9203-6	subsample	1160	0.048	0.029
268-8	subsample	1583	0.050	0.025	9203-7	subsample	1160	-0.011	0.029
268-9	subsample	791	0.095	0.036	9203-8	subsample	1160	0.081	0.029
285-1	headline	558	0.270	0.041	9203-9	baseline	540	0.087	0.043
285-10	subsample	238	0.397	0.061	9302-5	headline	2362	0.002	0.021
285-11	subsample	320	0.187	0.056	9305-1	headline	557	0.343	0.040
285-12	subsample	320	0.181	0.056	9305-10	robustness	219	0.290	0.066
285-2	baseline	567	0.239	0.041	9305-11	robustness	338	0.158	0.055
285-3	robustness	558	0.292	0.041	9305-2	baseline	557	0.336	0.040
285-4	robustness	558	0.239	0.041	9305-3	subsample	219	0.277	0.066
285-5	robustness	558	0.142	0.042	9305-4	subsample	338	0.224	0.054
285-6	robustness	558	0.112	0.042	9305-5	robustness	219	0.284	0.066
285-7	robustness	567	0.267	0.041	9305-6	robustness	338	0.206	0.054
285-8	robustness	567	0.206	0.041	9305-7	robustness	557	0.093	0.043
285-9	subsample	238	0.299	0.063	9305-8	robustness	219	0.130	0.069
302-1	headline	216	0.157	0.069	9305-9	robustness	338	0.110	0.055
323-1	headline	216	0.170	0.069	9401-1	headline	380	0.136	0.052

F Second-pass check

Table 16: Second-pass check by study: results confirmed (C), corrected (X) or unverifiable (U), heterogeneity entries corrected, and the second reader's view of the overall risk-of-bias rating.

ID	C	X	U	Het. X	RoB view
10	7	1	0	0	agree
29	8	0	0	0	agree
45	5	0	0	0	agree
57	5	1	0	0	too lenient
68	7	1	0	0	agree
71	3	0	0	0	agree
88	5	0	0	1	agree
108	6	0	0	1	agree
126	5	0	0	0	agree
138	9	0	0	1	agree
140	1	0	0	0	agree
162	6	0	0	0	agree
174	10	0	0	2	agree
217	6	0	0	0	agree
239	7	0	0	0	agree
245	6	0	0	0	agree
268	10	1	0	0	too harsh
269	3	0	0	0	agree
276	13	0	0	0	agree
278	5	0	0	0	agree
285	6	0	0	0	agree
302	3	0	0	0	too harsh
320	6	0	0	1	agree
321	6	1	0	0	agree
323	2	0	0	0	too harsh
341	6	0	0	0	agree
358	7	2	0	0	agree
392	3	1	0	0	agree
393	4	0	0	0	agree
407	6	0	0	0	agree
408	6	0	0	0	too harsh
422	5	1	0	0	agree
441	6	0	0	0	agree
442	0	2	1	0	agree
448	7	0	0	0	agree
535	8	0	0	0	agree
617	2	0	0	1	agree
707	7	0	0	1	too harsh
750	2	0	1	0	agree
755	9	0	0	0	agree
802	5	0	0	1	agree
803	7	1	0	0	agree
879	7	1	0	0	agree
891	6	0	0	1	agree
913	8	0	0	0	agree
952	10	0	0	0	agree
957	7	0	1	0	too lenient
988	3	0	0	1	agree
1013	3	0	0	0	agree
1025	4	0	0	0	agree
1026	4	1	0	0	agree
1031	9	1	0	1	too lenient
1192	4	0	0	0	agree
1372	3	0	0	0	agree
1419	6	0	0	0	agree
1428	4	2	0	0	agree
1438	4	0	0	0	agree
1448	4	0	0	0	agree
1469	8	0	0	0	too harsh

ID	C	X	U	Het. X	RoB view
1494	5	0	0	1	agree
1526	7	0	0	0	agree
1564	3	0	0	0	agree
1580	7	0	0	1	agree
1596	2	0	0	0	agree
1645	1	0	0	0	agree
1652	4	1	0	0	agree
1670	3	0	2	0	agree
1679	12	0	0	0	agree
1713	5	0	0	1	agree
1759	6	0	0	0	agree
1798	2	1	0	0	agree
9001	8	0	0	0	agree
9101	8	1	0	0	agree
9102	5	0	0	0	agree
9103	0	11	0	2	agree
9104	9	0	0	0	agree
9105	5	0	0	0	agree
9106	6	0	0	0	agree
9201	11	0	0	1	agree
9203	9	1	0	0	agree
9204	4	0	0	0	too lenient
9302	4	1	0	0	agree
9303	4	0	0	0	agree
9304	3	1	0	0	agree
9305	5	0	0	0	agree
9401	8	0	0	0	too harsh
9402	6	0	0	0	agree
9404	6	0	0	0	agree

G Macro-panel data and additional results

Variables

Table 17: World Bank WDI indicators used.

Role	Variable	WDI code
Technology	Mobile cellular subscriptions per 100 people	IT.CEL.SETS.P2
Technology	Individuals using the internet, % of population	IT.NET.USER.ZS
Technology	Fixed broadband subscriptions per 100 people	IT.NET.BBND.P2
Outcome	GDP per capita, constant 2015 US$	NY.GDP.PCAP.KD
Outcome	Life expectancy at birth	SP.DYN.LE00.IN
Outcome	Under-5 mortality rate	SH.DYN.MORT
Outcome	Unemployment, % of labour force	SL.UEM.TOTL.ZS
Outcome	Agriculture, value added % of GDP	NV.AGR.TOTL.ZS
Outcome	Electric power consumption per capita	EG.USE.ELEC.KH.PC
Control or moderator	School enrolment, secondary, gross	SE.SEC.ENRR
Control	Trade, % of GDP	NE.TRD.GNFS.ZS
Moderator	Domestic credit to private sector, % of GDP	FS.AST.PRVT.GD.ZS
Moderator	Access to electricity, % of population	EG.ELC.ACCS.ZS
Other	Population, total	SP.POP.TOTL

Income groups follow the World Bank's current classification of 217 economies (25 low, 47 lower-middle, 59 upper-middle and 86 high income; the analysis sample is smaller because of missing data). Aggregates were removed. CO_2 emissions per capita could not be retrieved from the API and are not analysed.

Table 18: Internet use and growth, same period (A1): change in annual growth (pp) per +10 points of users.

Income group	Economies	Observations	Effect of +10 on annual growth (pp)
Low	18	53	-0.916 [-3.241, 1.409] (p=0.44)
Lower-middle	42	158	1.185 [0.348, 2.023] (p=0.01)
Upper-middle	51	184	0.370 [-0.040, 0.779] (p=0.08)
High	60	223	-0.089 [-0.340, 0.162] (p=0.49)

Table 19: Internet use and growth, previous period (A2).

Income group	Economies	Observations	Effect of +10 on annual growth (pp)
Low	19	43	-3.861 [-7.810, 0.088] (p=0.06)
Lower-middle	41	127	-0.399 [-1.202, 0.405] (p=0.33)
Upper-middle	49	150	-0.209 [-0.582, 0.165] (p=0.27)
High	60	210	-0.306 [-0.509, -0.102] (p=0.00)

Table 20: Fixed broadband and growth, same period (A1). Not interpreted: extrapolation from few observations.

Income group	Economies	Observations	Effect of +10 on annual growth (pp)
Low	17	34	-21.426 [-27.860, -14.993] (p=0.00)
Lower-middle	40	99	5.846 [2.454, 9.237] (p=0.00)
Upper-middle	47	122	1.830 [0.623, 3.036] (p=0.00)
High	62	192	-0.057 [-0.589, 0.474] (p=0.83)

Table 21: Sensitivity: mobile adoption and growth, same period (A1), sample extended to 2023 (last period 2015–23).

Income group	Economies	Observations	Effect of +10 on annual growth (pp)
Low	20	60	0.329 [-0.123, 0.782] (p=0.15)
Lower-middle	42	166	0.344 [0.168, 0.521] (p=0.00)
Upper-middle	51	186	0.053 [-0.048, 0.154] (p=0.31)
High	61	227	0.134 [0.031, 0.238] (p=0.01)

Table 22: Sensitivity: previous-period specification (A2), sample extended to 2023.

Income group	Economies	Observations	Effect of +10 on annual growth (pp)
Low	19	54	-0.408 [-0.870, 0.054] (p=0.08)
Lower-middle	42	143	0.164 [-0.016, 0.344] (p=0.07)
Upper-middle	51	160	-0.047 [-0.195, 0.101] (p=0.53)
High	62	217	-0.187 [-0.302, -0.072] (p=0.00)

H Protocol amendments

1. (2 Oct 2026) The OpenAlex query was written with explicit phrase variants and restricted to the economics field, because an unrestricted query returned about 7,600 records, too many for one screener.
2. (2 Oct) Four holding categories were added at the abstract stage: context, environmental outcome only, reviews for backward citation search, and supplementary micro evidence.
3. (2 Oct) arXiv was attempted but not completed. Scopus, Web of Science and EconLit were not searched.
4. (2 Oct) Automated stage-1 rules were added before manual reading, with a single assisted screener and a verification sample of 90 decisions.

Table 23: Dose-response (A3): change in annual growth (pp) per +10 mobile subscriptions per 100, by baseline adoption level.

Sample	Baseline bin	Obs	Effect of +10 on annual growth (pp)
All economies	<10	186	0.405 [0.249, 0.561] (p=0.00)
All economies	10-40	97	0.281 [0.167, 0.395] (p=0.00)
All economies	40-80	123	0.179 [0.059, 0.298] (p=0.00)
All economies	>80	249	0.271 [0.119, 0.423] (p=0.00)
Low and lower-middle income	<10	109	0.509 [0.171, 0.846] (p=0.00)
Low and lower-middle income	10-40	36	0.359 [0.106, 0.612] (p=0.01)
Low and lower-middle income	40-80	53	0.453 [0.080, 0.826] (p=0.02)
Low and lower-middle income	>80	35	1.367 [-0.005, 2.739] (p=0.05)
Upper-middle and high income	<10	77	0.234 [0.013, 0.455] (p=0.04)
Upper-middle and high income	10-40	61	0.233 [0.093, 0.372] (p=0.00)
Upper-middle and high income	40-80	70	0.120 [-0.008, 0.247] (p=0.07)
Upper-middle and high income	>80	214	0.159 [0.052, 0.266] (p=0.00)

Table 24: Differences inside low and lower-middle income economies (A6).

Moderator (at start of period)	Interaction with +10 mobile (pp growth per SD)	Obs
secondary enrolment	0.136 [-0.205, 0.478] (p=0.43)	233
electricity access	0.024 [-0.301, 0.348] (p=0.89)	224
private credit	-0.074 [-0.295, 0.147] (p=0.51)	204
log GDP per capita	0.237 [-0.128, 0.602] (p=0.20)	233

Table 25: Event study with takeoff at 25 mobile subscriptions per 100 (log GDP per capita ×100; 400 bootstrap draws).

Sample	Treated	Pool	Pre-trend: mean of effects at -5..-2 [95% CI]	Passes pre-trend rule	Effect at
All economies	188	188	-0.2 [-0.7, 0.3]	yes	6.1 [1.8,
Low and lower-middle income	66	188	0.2 [-0.7, 1.4]	yes	-0.5 [-6.1
Upper-middle income	56	188	-0.9 [-2.3, 0.6]	yes	7.5 [-0.5
High income	66	188	0.0 [-1.1, 0.6]	yes	9.3 [3.9,

Table 26: Event study with takeoff at 50 mobile subscriptions per 100. Samples that fail the pre-trend rule are not interpreted.

Sample	Treated	Pool	Pre-trend: mean of effects at -5..-2 [95% CI]	Passes pre-trend rule	Effect at
All economies	176	188	-0.9 [-1.4, -0.3]	no	4.6 [-0.1
Low and lower-middle income	57	188	-0.3 [-1.6, 1.0]	yes	2.1 [-6.6
Upper-middle income	53	188	-1.9 [-3.5, -0.3]	no	5.3 [-1.6
High income	66	188	-0.6 [-1.5, 0.2]	yes	4.9 [0.2,

5. (2 Oct) Included records were ranked in two tiers to decide which paywalled studies to request first.
6. (2 Oct) Records with no free full text and not on the short request list were kept as an abstract-level evidence map, excluded from pooled estimates and from risk-of-bias ratings.
7. (3 Oct) Studies whose only available venue the author judged unreliable were not read and not used.
8. (3 Oct) Four studies (298, 474, 878 and 1495) were moved from include to exclude after a second-pass check. Five regression estimates were excluded from the meta-regression after the same check.
9. (4 Oct) A blind re-screening of a random sample flagged missed records; five were obtained and read (448, 474, 755, 891, 9001), four were included and 474 was moved to exclude.
10. (4 Oct) Because the re-screening and then a first-page check of one Google Scholar query showed that the search had missed studies, the supplementary Google Scholar queries were added (G1 to

Table 27: Event study, secondary outcomes (takeoff at 10 per 100): effects at five and ten years. Rows marked fails do not pass the pre-trend rule and are not interpreted.

Outcome	Sample	Treated	Pre-trend mean [95% CI]	Effect at +5
Life expectancy (years)	All economies	178	0.05 [-0.00, 0.08] (passes)	0.86 [0.10, 1.69
Life expectancy (years)	Low and lower-middle income	66	0.03 [-0.07, 0.11] (passes)	2.61 [-0.49, 6.3
Life expectancy (years)	Upper-middle income	56	0.05 [-0.03, 0.13] (passes)	0.63 [-0.30, 1.7
Life expectancy (years)	High income	56	0.02 [-0.08, 0.18] (passes)	0.81 [0.02, 1.54
Under-5 mortality (log x 100)	All economies	178	-0.09 [-0.35, 0.28] (passes)	-4.71 [-13.90, 1
Under-5 mortality (log x 100)	Low and lower-middle income	66	-0.20 [-0.72, 0.54] (passes)	-15.08 [-26.03,
Under-5 mortality (log x 100)	Upper-middle income	56	0.02 [-0.57, 0.53] (passes)	-2.97 [-21.71, 6
Under-5 mortality (log x 100)	High income	56	0.21 [-0.47, 0.65] (passes)	-4.01 [-12.47, 4
Unemployment (pp)	All economies	178	0.11 [0.01, 0.21] (fails)	-0.22 [-1.24, 0.
Unemployment (pp)	Low and lower-middle income	66	-0.16 [-0.30, -0.03] (fails)	n/a
Unemployment (pp)	Upper-middle income	56	0.54 [0.23, 0.74] (fails)	1.34 [-1.33, 2.9
Unemployment (pp)	High income	56	0.01 [-0.27, 0.37] (passes)	-1.14 [-2.34, -0
Agriculture share of value added (pp)	All economies	178	0.27 [0.02, 0.62] (fails)	-3.56 [-6.14, -1
Agriculture share of value added (pp)	Low and lower-middle income	66	0.34 [-0.22, 1.05] (passes)	n/a
Agriculture share of value added (pp)	Upper-middle income	56	0.36 [-0.05, 0.74] (passes)	-4.29 [-8.59, -0
Agriculture share of value added (pp)	High income	56	0.16 [-0.19, 0.52] (passes)	-3.24 [-5.71, -1
Electric power use per capita (log x 100)	All economies	178	0.01 [-1.09, 1.04] (passes)	10.26 [-3.84, 2
Electric power use per capita (log x 100)	Low and lower-middle income	66	-0.44 [-2.49, 1.76] (passes)	n/a
Electric power use per capita (log x 100)	Upper-middle income	56	-0.46 [-1.98, 2.11] (passes)	2.12 [-17.66, 2
Electric power use per capita (log x 100)	High income	56	0.43 [-0.98, 1.21] (passes)	14.04 [-1.25, 2

G15; the four run are listed in Appendix C). Only the first pages of G1, G2, G4 and G7 have been run. Twenty-nine of their 40 results were new; twenty full texts were obtained and read (9101 to 9106, 9201 to 9204, 9301 to 9305, 9401 to 9404 and 1580), seventeen were included and three (9202, five countries; 9301, a model without estimation; 9403, estimated on a simulated panel) excluded. Pilot meta-regression, exploratory analyses and all counts were rerun after their inclusion, and the earlier moderator results were superseded. Further amendment (4 Oct): a G2 result removed by the automated stage-1 rule (Yin and Choi 2023) was recovered, showing that the stage-1 term list was too narrow.

11. (3 Oct) The event-study design in the macro-panel plan was replaced by a staggered design with not-yet-treated controls and covariate adjustment after the first design failed its pre-trend checks; local projections were added.

I PRISMA 2020 checklist

Table 28: PRISMA 2020 checklist: where each item is reported. Section numbers refer to this paper.

Item	Topic	Requirement	Reported	Where
1	Title	Identifies the report as a systematic review	Yes	Title
2	Abstract	Structured abstract	Partly	Abstract (not the full PRISMA abstract checklist)
3	Rationale	Rationale in context of existing knowledge	Yes	Introduction; Conceptual background
4	Objectives	Explicit objectives and questions	Yes	Introduction; Conceptual background
5	Eligibility criteria	Inclusion and exclusion criteria	Yes	Methods (eligibility); Appendix D
6	Information sources	Databases, dates, last search	Partly	Methods (sources); Scopus, Web of Science, EconLit not searched; Google Scholar partly
7	Search strategy	Full strategy for all sources	Yes	Appendix C

Item	Topic	Requirement	Reported	Where
8	Selection process	How records were screened, number of reviewers	Partly	Methods (screening); one assisted reviewer plus a blind model re-screen of a random sample (Results, cross-checks)
9	Data collection process	Methods of extraction, number of reviewers	Partly	Methods (extraction, second pass); machine extraction, machine second pass, blind re-extraction of 16 studies, author hand check of a random sample
10	Data items	Outcomes and other variables	Yes	Methods (eligibility); Appendix D
11	Risk of bias	Tools and reviewers	Partly	Methods (risk of bias); adapted tool, single rater
12	Effect measures	Measures used for each outcome	Yes	Methods (meta-regression)
13a–f	Synthesis methods	Eligibility for synthesis, methods, heterogeneity, sensitivity	Partly	Methods (synthesis, meta-regression); leave-one-out and quality-restricted analyses in Results (meta-regression)
14	Reporting bias assessment	Methods to assess reporting bias	Yes	Methods and Results (meta-regression: FAT-PET, PEESE)
15	Certainty assessment	Methods to assess certainty (e.g. GRADE)	No	Not done; listed as a limitation
16	Study selection results	Flow diagram, excluded studies	Yes	Results (study selection), Figure 1; reasons coded
17	Study characteristics	Cite and present characteristics	Yes	Results (characteristics), Appendix A
18	Risk of bias in studies	Assessments for each study	Yes	Results (risk of bias), Appendix A
19	Results of individual studies	Summary and effect estimates	Partly	Appendices B and E; no forest plot for all outcomes
20	Results of syntheses	Results, heterogeneity, sensitivity	Yes	Results (all synthesis subsections)
21	Reporting biases	Assessments of reporting bias	Yes	Results (meta-regression)
22	Certainty of evidence	Certainty for each outcome	No	Not done
23	Discussion	Interpretation, limitations, implications	Yes	Discussion; Limitations
24	Registration and protocol	Registration number, protocol access	Partly	Protocol available with the data and code; not registered
25	Support	Funding and sponsors	Yes	Declarations (author to confirm)
26	Competing interests	Declared	No	Author to complete
27	Availability of data and code	Public availability	Yes	Declarations